

WHITNEY FORMS

IZ TECÚM

ABSTRACT. SLMath notes.

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Part 1. Partial differential equations

1. ELLIPTIC EQUATIONS

Throughout, $\Omega \subset \mathbb{R}^d$ is bounded, connected, and Lipschitz; all coefficients, data, and function spaces are real.

1.1. Strong elliptic problems.

Definition 1.1.1 (Partial derivatives, gradient, and Jacobian). Let $u \in C^2(\Omega)$ and $\mathbf{F} = [F^1, \dots, F^d]^\top \in C^1(\Omega; \mathbb{R}^d)$ [4, Chapter 2 and Appendix C]. Fix $\mathbf{x} \in \Omega$ and $1 \leq j \leq d$.

$$\partial_{x_j}(u)(\mathbf{x}) = \lim_{t \rightarrow 0} \frac{u(\mathbf{x} + t\mathbf{e}_j) - u(\mathbf{x})}{t}, \quad \nabla u = \begin{bmatrix} \partial_{x_1}(u) \\ \vdots \\ \partial_{x_d}(u) \end{bmatrix}, \quad D\mathbf{F} = \begin{bmatrix} \partial_{x_1}(F^1) & \cdots & \partial_{x_d}(F^1) \\ \vdots & \ddots & \vdots \\ \partial_{x_1}(F^d) & \cdots & \partial_{x_d}(F^d) \end{bmatrix}.$$

Definition 1.1.2 (Divergence, Hessian, and Laplacian). For u and $\mathbf{F}$ as in Definition 1.1.1,

$$\nabla \cdot \mathbf{F} = [\partial_{x_1} \quad \cdots \quad \partial_{x_d}] \cdot \begin{bmatrix} F^1 \\ \vdots \\ F^d \end{bmatrix} = \sum_{j=1}^d \partial_{x_j}(F^j) = \text{tr}(D\mathbf{F}),$$

$$D^2u = \begin{bmatrix} \partial_{x_1}^2(u) & \cdots & \partial_{x_1}\partial_{x_d}(u) \\ \vdots & \ddots & \vdots \\ \partial_{x_d}\partial_{x_1}(u) & \cdots & \partial_{x_d}^2(u) \end{bmatrix}, \quad \Delta u = \sum_{j=1}^d \partial_{x_j}^2(u) = \text{tr}(D^2u) = \nabla \cdot (\nabla u).$$

Definition 1.1.3 (Poisson–Dirichlet problem). For $f : \Omega \rightarrow \mathbb{R}$ and $g : \partial\Omega \rightarrow \mathbb{R}$, the Poisson–Dirichlet problem is [1, Sections 1.1–1.2]

$$\begin{cases} -\Delta u = f, & \mathbf{x} \in \Omega, \\ u = g, & \mathbf{x} \in \partial\Omega, \end{cases} \quad \xrightarrow{1.1.2} \quad -\Delta u = -[\partial_{x_1} \quad \cdots \quad \partial_{x_d}] \cdot \begin{bmatrix} \partial_{x_1}(u) \\ \vdots \\ \partial_{x_d}(u) \end{bmatrix} = -\sum_{j=1}^d \partial_{x_j}^2(u),$$

$$B_1(\mathbf{0}) = \{\mathbf{x} \in \mathbb{R}^d : |\mathbf{x}| < 1\}, \quad \partial B_1(\mathbf{0}) = \{\mathbf{x} \in \mathbb{R}^d : |\mathbf{x}| = 1\}.$$

Definition 1.1.4 (Uniformly elliptic diffusion tensor). Let $A = [a^{ij}]_{i,j=1}^d \in L^\infty(\Omega; \mathbb{R}^{d \times d})$ be symmetric a.e., with

$$Q_A(\mathbf{x}, \xi) := [\xi_1 \quad \cdots \quad \xi_d] \cdot \begin{bmatrix} a^{11} & \cdots & a^{1d} \\ \vdots & \ddots & \vdots \\ a^{d1} & \cdots & a^{dd} \end{bmatrix} \cdot \begin{bmatrix} \xi_1 \\ \vdots \\ \xi_d \end{bmatrix} = \sum_{i=1}^d \sum_{j=1}^d a^{ij}(\mathbf{x}) \xi_i \xi_j,$$

$$0 < \lambda_A \leq \Lambda_A < \infty, \quad \lambda_A \sum_{i=1}^d \xi_i^2 \leq Q_A(\mathbf{x}, \xi) \leq \Lambda_A \sum_{i=1}^d \xi_i^2 \quad (\text{a.e. } \mathbf{x} \in \Omega, \xi \in \mathbb{R}^d).$$

Definition 1.1.5 (Diffusion flux and conormal derivative). For A as in Definition 1.1.4 and u as in Definition 1.1.1,

$$A \cdot \nabla u = \begin{bmatrix} a^{11} & \cdots & a^{1d} \\ \vdots & \ddots & \vdots \\ a^{d1} & \cdots & a^{dd} \end{bmatrix} \cdot \begin{bmatrix} \partial_{x_1}(u) \\ \vdots \\ \partial_{x_d}(u) \end{bmatrix} = \begin{bmatrix} \sum_{j=1}^d a^{1j} \partial_{x_j}(u) \\ \vdots \\ \sum_{j=1}^d a^{dj} \partial_{x_j}(u) \end{bmatrix},$$

$$\begin{aligned} \xrightarrow{1.1.2} \nabla \cdot (A \cdot \nabla u) &= \sum_{i=1}^d \partial_{x_i} \left(\sum_{j=1}^d a^{ij} \partial_{x_j}(u) \right) = \sum_{i=1}^d \sum_{j=1}^d \partial_{x_i} (a^{ij} \partial_{x_j}(u)), \\ \mathbf{n}^\top \cdot A \cdot \nabla u &= [n^1 \quad \dots \quad n^d] \cdot \begin{bmatrix} a^{11} & \dots & a^{1d} \\ \vdots & \ddots & \vdots \\ a^{d1} & \dots & a^{dd} \end{bmatrix} \cdot \begin{bmatrix} \partial_{x_1}(u) \\ \vdots \\ \partial_{x_d}(u) \end{bmatrix} = \sum_{i=1}^d \sum_{j=1}^d n^i a^{ij} \partial_{x_j}(u). \end{aligned}$$

The expansions are pointwise for $A \in C^1(\overline{\Omega}; \mathbb{R}^{d \times d})$, $u \in C^2(\overline{\Omega})$. For $u \in H^1(\Omega)$ and $\mathbf{q} = A \cdot \nabla u \in H(\nabla \cdot, \Omega)$,

$$\nabla \cdot \mathbf{q} \in L^2(\Omega), \quad \mathbf{n}^\top \cdot \mathbf{q} \in H^{-1/2}(\partial\Omega).$$

Definition 1.1.6 (Mixed boundary partition). Let $\Gamma_D, \Gamma_N, \Gamma_R \subset \partial\Omega$ be relatively open with Lipschitz relative interfaces and

$$\partial\Omega = \overline{\Gamma_D} \cup \overline{\Gamma_N} \cup \overline{\Gamma_R}, \quad \Gamma_D \cap \Gamma_N = \Gamma_D \cap \Gamma_R = \Gamma_N \cap \Gamma_R = \emptyset, \quad |\Gamma_D|_{d-1} > 0.$$

Definition 1.1.7 (Diffusion–reaction problem with mixed boundary data). For A as in Definition 1.1.4, fluxes as in Definition 1.1.5, $\Gamma_D, \Gamma_N, \Gamma_R$ as in Definition 1.1.6, $c \in L^\infty(\Omega)$, $\alpha \in L^\infty(\Gamma_R)$, and $c, \alpha \geq 0$, the mixed problem is [1, 2, Sections 2.1–2.6]

$$\begin{cases} -\nabla \cdot (A \cdot \nabla u) + c \cdot u = f, & \mathbf{x} \in \Omega, \\ u = g_D, & \mathbf{x} \in \Gamma_D, \\ \mathbf{n}^\top \cdot A \cdot \nabla u = g_N, & \mathbf{x} \in \Gamma_N, \\ \mathbf{n}^\top \cdot A \cdot \nabla u + \alpha \cdot u = g_R, & \mathbf{x} \in \Gamma_R, \end{cases}$$

$$\Gamma_R = \emptyset \xrightarrow{1.1.6} \partial\Omega = \overline{\Gamma_D} \cup \overline{\Gamma_N}, \quad \Gamma_N = \Gamma_R = \emptyset \xrightarrow{1.1.6} \partial\Omega = \overline{\Gamma_D}.$$

Corollary 1.1.8 (Poisson as the scalar diffusion problem). If $A = I_d$, $c = 0$, $\Gamma_D = \partial\Omega$, and $\Gamma_N = \Gamma_R = \emptyset$, then the mixed problem of Definition 1.1.7 is the Poisson–Dirichlet problem of Definition 1.1.3 with $g = g_D$.

Proof. Let $A = I_d$, $c = 0$, and $\Gamma_D = \partial\Omega$; the flux reduces to the gradient, the reaction term vanishes, and the three boundary conditions collapse to one Dirichlet condition.

$$\xrightarrow{1.1.5} I_d \cdot \begin{bmatrix} \partial_{x_1}(u) \\ \vdots \\ \partial_{x_d}(u) \end{bmatrix} = \begin{bmatrix} \partial_{x_1}(u) \\ \vdots \\ \partial_{x_d}(u) \end{bmatrix}, \quad c = 0 \implies c \cdot u = 0, \quad \Gamma_N = \Gamma_R = \emptyset \xrightarrow{1.1.6} \partial\Omega = \overline{\Gamma_D},$$

$$\xrightarrow{1.1.2} -[\partial_{x_1} \quad \dots \quad \partial_{x_d}] \cdot I_d \cdot \begin{bmatrix} \partial_{x_1}(u) \\ \vdots \\ \partial_{x_d}(u) \end{bmatrix} = -\sum_{j=1}^d \partial_{x_j}^2(u) = -\Delta u,$$

$$(A, c, \Gamma_D, \Gamma_N, \Gamma_R) = (I_d, 0, \partial\Omega, \emptyset, \emptyset) \xrightarrow{1.1.7} \begin{cases} -\Delta u = f, & \mathbf{x} \in \Omega, \\ u = g_D, & \mathbf{x} \in \partial\Omega, \end{cases}$$

$$g = g_D \xleftrightarrow{1.1.3} \begin{cases} -\Delta u = f, & \mathbf{x} \in \Omega, \\ u = g, & \mathbf{x} \in \partial\Omega. \end{cases}$$

□

Example 1.1.9 (An interior Dirichlet region). Let $D \Subset \Omega$ have smooth boundary, let $\Omega_D = \Omega \setminus \bar{D}$ be connected and Lipschitz, and set $(A, c, \Gamma_D, \Gamma_N, \Gamma_R) = (I_d, 0, \partial D, \partial \Omega, \emptyset)$, with $\mathbf{n}$ outward for Ω_D .

$$A = I_d \xrightarrow{1.1.5} \mathbf{n}^\top \cdot \nabla u = [n^1 \ \cdots \ n^d] \cdot \begin{bmatrix} \partial_{x_1}(u) \\ \vdots \\ \partial_{x_d}(u) \end{bmatrix} = \sum_{j=1}^d n^j \partial_{x_j}(u),$$

$$(I_d, 0, \partial D, \partial \Omega, \emptyset) \xrightarrow{1.1.7} \begin{cases} -\Delta u = f, & \mathbf{x} \in \Omega_D, \\ u = g_D, & \mathbf{x} \in \partial D, \\ \mathbf{n}^\top \cdot \nabla u = g_N, & \mathbf{x} \in \partial \Omega. \end{cases}$$

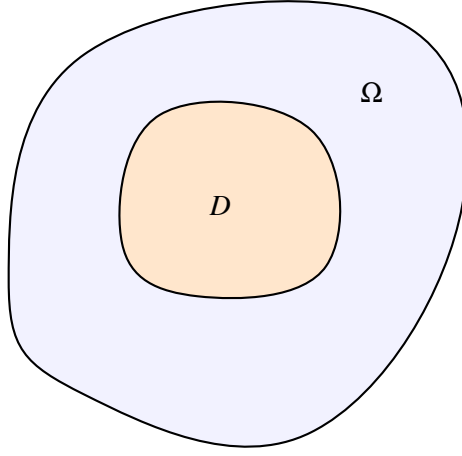


FIGURE 1. The Dirichlet region $D \Subset \Omega$, with $\Gamma_D = \partial D$ and $\Gamma_N = \partial \Omega$.

Example 1.1.10 (Stationary heat equation). For $0 < t < T$, let $A, c, \alpha, F, g_D, g_N, g_R$ be time independent and let y satisfy [4, Chapter 2]

$$\begin{cases} \partial_t(y) - \nabla \cdot (A \cdot \nabla y) + c \cdot y = F, & (\mathbf{x}, t) \in \Omega \times (0, T), \\ y = g_D, & (\mathbf{x}, t) \in \Gamma_D \times (0, T), \\ \mathbf{n}^\top \cdot A \cdot \nabla y = g_N, & (\mathbf{x}, t) \in \Gamma_N \times (0, T), \\ \mathbf{n}^\top \cdot A \cdot \nabla y + \alpha \cdot y = g_R, & (\mathbf{x}, t) \in \Gamma_R \times (0, T), \end{cases}$$

$$\partial_t(y) = 0 \xrightarrow{1.1.7} \begin{cases} -\nabla \cdot (A \cdot \nabla y) + c \cdot y = F, & \mathbf{x} \in \Omega, \\ y = g_D, & \mathbf{x} \in \Gamma_D, \\ \mathbf{n}^\top \cdot A \cdot \nabla y = g_N, & \mathbf{x} \in \Gamma_N, \\ \mathbf{n}^\top \cdot A \cdot \nabla y + \alpha \cdot y = g_R, & \mathbf{x} \in \Gamma_R. \end{cases}$$

Fact 1.1.11 (Fundamental theorem of calculus). If $w \in C^1([a, b])$, then

$$w(x) - w(a) = \int_a^x w'(s) ds \quad (a \leq x \leq b).$$

Proof. [4, Appendix C]. □

Example 1.1.12 (The two-point source problem). Consider

$$\begin{aligned} & \begin{cases} -u''(x) = x^2, & 0 < x < 1, \\ u(0) = 0, \\ u(1) = 1, \end{cases} \quad \xLeftrightarrow{1.1.3} (d, f, g) = (1, x^2, x). \\ & u''(x) = -x^2 \xRightarrow{1.1.11} u'(x) - u'(0) = \int_0^x u''(s) ds = -\int_0^x s^2 ds = -\frac{x^3}{3}, \\ & u'(x) = u'(0) - \frac{x^3}{3} \xRightarrow{1.1.11} u(x) - u(0) = \int_0^x \left(u'(0) - \frac{t^3}{3} \right) dt = x \cdot u'(0) - \frac{x^4}{12}, \\ & u(0) = 0, \quad u(1) = 1 \implies u'(0) - \frac{1}{12} = 1 \implies u'(0) = \frac{13}{12}, \\ & u(x) = \frac{13}{12}x - \frac{1}{12}x^4, \quad -u''(x) = x^2, \quad (u(0), u(1)) = (0, 1). \end{aligned}$$

Example 1.1.13 (Laplacian and divergence computations). Let

$$\begin{aligned} & v(x_1, x_2) = x_1^7 \sin(x_2), \quad \tilde{v}(x_1, x_2) = 7x_1^6 \sin(x_2), \quad w(x_1, x_2) = \cos(x_1)x_2^7. \\ & \xRightarrow{1.1.1} \nabla v = \begin{bmatrix} 7x_1^6 \sin(x_2) \\ x_1^7 \cos(x_2) \end{bmatrix}, \quad D^2 v = \begin{bmatrix} 42x_1^5 \sin(x_2) & 7x_1^6 \cos(x_2) \\ 7x_1^6 \cos(x_2) & -x_1^7 \sin(x_2) \end{bmatrix}, \\ & \xRightarrow{1.1.2} \Delta v = \text{tr}(D^2 v) = 42x_1^5 \sin(x_2) - x_1^7 \sin(x_2) = x_1^5(42 - x_1^2) \sin(x_2), \\ & D^2 \tilde{v} = \begin{bmatrix} 210x_1^4 \sin(x_2) & 42x_1^5 \cos(x_2) \\ 42x_1^5 \cos(x_2) & -7x_1^6 \sin(x_2) \end{bmatrix}, \quad \Delta \tilde{v} = 210x_1^4 \sin(x_2) - 7x_1^6 \sin(x_2), \\ & \xRightarrow{1.1.1} \nabla w = \begin{bmatrix} \partial_{x_1}(\cos(x_1)x_2^7) \\ \partial_{x_2}(\cos(x_1)x_2^7) \end{bmatrix} = \begin{bmatrix} -x_2^7 \sin(x_1) \\ 7x_2^6 \cos(x_1) \end{bmatrix}. \end{aligned}$$

For $\mathbf{F}(x_1, x_2, x_3) = [x_1 x_2, x_3 \cos(x_2), x_2 \sin(x_3)]^T$,

$$\xRightarrow{1.1.1} D\mathbf{F} = \begin{bmatrix} x_2 & x_1 & 0 \\ 0 & -x_3 \sin(x_2) & \cos(x_2) \\ 0 & \sin(x_3) & x_2 \cos(x_3) \end{bmatrix},$$

$$\xRightarrow{1.1.2} \nabla \cdot \mathbf{F} = \partial_{x_1}(x_1 x_2) + \partial_{x_2}(x_3 \cos(x_2)) + \partial_{x_3}(x_2 \sin(x_3)) = x_2 - x_3 \sin(x_2) + x_2 \cos(x_3) = \text{tr}(D\mathbf{F}).$$

Definition 1.1.14 (Dirichlet Laplacian eigenpair). For $\lambda \in \mathbb{R}$ and $v \in C^2(\Omega) \cap C(\overline{\Omega})$, set [4, Section 6.5]

$$\mathcal{E}_D(\Omega) := \{(\lambda, v) \in \mathbb{R} \times (C^2(\Omega) \cap C(\overline{\Omega})) : v \neq 0, -\Delta v = \lambda v, v|_{\partial\Omega} = 0\}.$$

Example 1.1.15 (A rectangular Dirichlet eigenmode). Let $m, n \in \{1, 2, \dots\}$, $\Omega = (0, 2\pi)^2$, $\lambda := n^2 + m^2$, and $v(x_1, x_2) = \sin(nx_1) \sin(mx_2)$.

$$\begin{aligned} & \xRightarrow{1.1.1} \nabla v = \begin{bmatrix} n \cos(nx_1) \sin(mx_2) \\ m \sin(nx_1) \cos(mx_2) \end{bmatrix}, \quad D^2 v = \begin{bmatrix} -n^2 v & nm \cos(nx_1) \cos(mx_2) \\ nm \cos(nx_1) \cos(mx_2) & -m^2 v \end{bmatrix}, \\ & \xRightarrow{1.1.2} -\Delta v = -\text{tr}(D^2 v) = (n^2 + m^2)v = \lambda v, \quad v\left(\frac{\pi}{2n}, \frac{\pi}{2m}\right) = 1 \neq 0, \\ & v(0, x_2) = v(2\pi, x_2) = v(x_1, 0) = v(x_1, 2\pi) = 0 \implies v|_{\partial\Omega} = 0, \\ & v \neq 0, \quad -\Delta v = \lambda v, \quad v|_{\partial\Omega} = 0 \xLeftrightarrow{1.1.14} (\lambda, v) \in \mathcal{E}_D(\Omega). \end{aligned}$$

1.2. Green's identity.

Theorem 1.2.1 (Divergence theorem). *If $\mathbf{G} \in C^1(\overline{\Omega}; \mathbb{R}^d)$, then*

$$\int_{\Omega} \nabla \cdot \mathbf{G} \, d\mathbf{x} = \int_{\partial\Omega} \mathbf{n}^T \cdot \mathbf{G} \, ds.$$

Proof. [4, Appendix C.2]. □

Proposition 1.2.2 (Weighted divergence identity). *If $\omega \in C^1(\overline{\Omega})$ and $\mathbf{G} \in C^1(\overline{\Omega}; \mathbb{R}^d)$, then*

$$\int_{\Omega} \omega \cdot \nabla \cdot \mathbf{G} \, d\mathbf{x} = \int_{\partial\Omega} \omega \cdot \mathbf{n}^T \cdot \mathbf{G} \, ds - \int_{\Omega} (\nabla \omega)^T \cdot \mathbf{G} \, d\mathbf{x}.$$

Proof. Let $\omega \mathbf{G} \in C^1(\overline{\Omega}; \mathbb{R}^d)$; the product rule splits its divergence into a transport term and a weighted divergence, and the total divergence integrates to a boundary flux.

$$\begin{aligned} \stackrel{1.1,2}{\implies} \nabla \cdot (\omega \mathbf{G}) &= \sum_{j=1}^d \partial_{x_j}(\omega G^j) = \sum_{j=1}^d \partial_{x_j}(\omega) G^j + \sum_{j=1}^d \omega \partial_{x_j}(G^j) = (\nabla \omega)^T \cdot \mathbf{G} + \omega \cdot \nabla \cdot \mathbf{G}, \\ \omega \mathbf{G} \in C^1(\overline{\Omega}; \mathbb{R}^d) &\stackrel{1.2,1}{\implies} \int_{\Omega} (\nabla \omega)^T \cdot \mathbf{G} \, d\mathbf{x} + \int_{\Omega} \omega \cdot \nabla \cdot \mathbf{G} \, d\mathbf{x} = \int_{\partial\Omega} \omega \cdot \mathbf{n}^T \cdot \mathbf{G} \, ds, \\ \int_{\Omega} \omega \cdot \nabla \cdot \mathbf{G} \, d\mathbf{x} &= \int_{\partial\Omega} \omega \cdot \mathbf{n}^T \cdot \mathbf{G} \, ds - \int_{\Omega} (\nabla \omega)^T \cdot \mathbf{G} \, d\mathbf{x} \quad [1, \text{Sections 2.1--2.2}]. \end{aligned}$$

□

Definition 1.2.3 (Energy test space). Let $\mathcal{H}_D$ denote the trace on Γ_D of Definition 1.1.6, and set [2, Sections 1.7 and 2.1]

$$V = \overline{\{\varphi \in C^\infty(\overline{\Omega}) : \varphi|_{\Gamma_D} = 0\}}^{H^1} = \{v \in H^1(\Omega) : \mathcal{H}_D(v) = 0\}, \quad \|v\|_V = \|v\|_{H^1(\Omega)}.$$

Definition 1.2.4 (Energy form, load functional, and Dirichlet lift). For the data of Definition 1.1.7, $\mathcal{H}_D(G) = g_D$, $u = G + w$, and $w \in V$, define

$$\begin{aligned} a(u, v) &= \int_{\Omega} (\nabla v)^T \cdot \mathbf{A} \cdot \nabla u \, d\mathbf{x} + \int_{\Omega} c \cdot u \cdot v \, d\mathbf{x} + \int_{\Gamma_R} \alpha \cdot u \cdot v \, ds, \\ \ell(v) &= \int_{\Omega} f \cdot v \, d\mathbf{x} + \int_{\Gamma_N} g_N \cdot v \, ds + \int_{\Gamma_R} g_R \cdot v \, ds, \quad a(w, v) = \ell(v) - a(G, v) \quad (v \in V). \end{aligned}$$

Definition 1.2.5 (Weak mixed solution).

$$\mathcal{W}_{\text{mix}} := \{u \in V : a(u, v) = \ell(v) \ (v \in V)\}.$$

Fact 1.2.6 (Green's identity). *For smooth v and $\mathbf{q}$,*

$$\int_{\Omega} -\nabla \cdot \mathbf{q} \cdot v \, d\mathbf{x} = \int_{\Omega} (\nabla v)^T \cdot \mathbf{q} \, d\mathbf{x} - \int_{\partial\Omega} (\mathbf{n}^T \cdot \mathbf{q}) \cdot v \, ds.$$

Proof. Let $(\omega, \mathbf{G}) = (v, \mathbf{q})$; integration by parts moves the divergence onto the test function.

$$(\omega, \mathbf{G}) = (v, \mathbf{q}) \stackrel{1.2,2}{\implies} \int_{\Omega} -\nabla \cdot \mathbf{q} \cdot v \, d\mathbf{x} = \int_{\Omega} (\nabla v)^T \cdot \mathbf{q} \, d\mathbf{x} - \int_{\partial\Omega} (\mathbf{n}^T \cdot \mathbf{q}) \cdot v \, ds.$$

□

Corollary 1.2.7 (Weak Poisson equation). *If $g = 0$, $u \in C^2(\overline{\Omega})$, and u satisfies Definition 1.1.3, then*

$$\int_{\Omega} (\nabla v)^T \cdot \nabla u \, d\mathbf{x} = \int_{\Omega} f \cdot v \, d\mathbf{x} \quad (v \in H_0^1(\Omega)).$$

Proof. Let $v \in C_c^\infty(\Omega)$ and $\mathbf{q} = \nabla u$; testing the strong equation and integrating by parts moves one derivative onto v , and the compact support kills the boundary flux.

$$\begin{aligned} -\Delta u &= f \xrightarrow{1.1,2} \int_{\Omega} f \cdot v \, d\mathbf{x} = - \int_{\Omega} \nabla \cdot (\nabla u) \cdot v \, d\mathbf{x} \quad (v \in C_c^\infty(\Omega)), \\ (\mathbf{q} = \nabla u, v|_{\partial\Omega} = 0) &\xrightarrow{1.2,6} - \int_{\Omega} \nabla \cdot (\nabla u) \cdot v \, d\mathbf{x} = \int_{\Omega} (\nabla v)^\top \cdot \nabla u \, d\mathbf{x} - \underbrace{\int_{\partial\Omega} v \cdot \mathbf{n}^\top \cdot \nabla u \, ds}_0. \end{aligned}$$

Both sides are H^1 -continuous in v , and $C_c^\infty(\Omega)$ is dense in $H_0^1(\Omega)$.

$$\begin{aligned} L_u(v) &:= \int_{\Omega} (\nabla v)^\top \cdot \nabla u \, d\mathbf{x} - \int_{\Omega} f \cdot v \, d\mathbf{x}, \quad L_u|_{C_c^\infty(\Omega)} = 0, \\ |L_u(v)| &\leq \left(\|\nabla u\|_{L^2(\Omega)} + \|f\|_{L^2(\Omega)} \right) \|v\|_{H^1(\Omega)} \implies L_u \in (H_0^1(\Omega))^*, \\ \left(L_u \in (H_0^1(\Omega))^*, L_u|_{C_c^\infty(\Omega)} = 0, \overline{C_c^\infty(\Omega)}^{H^1} = H_0^1(\Omega) \right) &\implies L_u|_{H_0^1(\Omega)} = 0, \\ L_u|_{H_0^1(\Omega)} = 0 &\implies \int_{\Omega} (\nabla v)^\top \cdot \nabla u \, d\mathbf{x} = \int_{\Omega} f \cdot v \, d\mathbf{x} \quad (v \in H_0^1(\Omega)). \end{aligned}$$

□

Proposition 1.2.8 (Strong-to-weak reduction). *If $u \in H^1(\Omega)$, $A \cdot \nabla u \in C^1(\overline{\Omega}; \mathbb{R}^d)$, and u satisfies Definition 1.1.7 with $g_D = 0$, then u is a weak mixed solution.*

Proof. Let $\mathbf{q} = A \cdot \nabla u$ and $v \in C^\infty(\overline{\Omega}) \cap V$; testing the strong equation splits the boundary flux along $\Gamma_D \cup \Gamma_N \cup \Gamma_R$, and each piece feeds one term of a or ℓ .

$$\begin{aligned} (u \in H^1(\Omega), u|_{\Gamma_D} = g_D = 0) &\xrightarrow{1.2,3} u \in V, \quad -\nabla \cdot \mathbf{q} + c \cdot u = f, \\ \mathbf{q} = A \cdot \nabla u &\xrightarrow{1.2,6} \int_{\Omega} f \cdot v \, d\mathbf{x} = \int_{\Omega} (\nabla v)^\top \cdot A \cdot \nabla u \, d\mathbf{x} + \int_{\Omega} c \cdot u \cdot v \, d\mathbf{x} - \int_{\partial\Omega} v \cdot \mathbf{n}^\top \cdot \mathbf{q} \, ds, \\ v|_{\Gamma_D} = 0 &\xrightarrow{1.1,6} \int_{\partial\Omega} v \cdot \mathbf{n}^\top \cdot \mathbf{q} \, ds = \int_{\Gamma_N} v \cdot \mathbf{n}^\top \cdot \mathbf{q} \, ds + \int_{\Gamma_R} v \cdot \mathbf{n}^\top \cdot \mathbf{q} \, ds, \\ (\mathbf{n}^\top \cdot \mathbf{q}|_{\Gamma_N} = g_N, \mathbf{n}^\top \cdot \mathbf{q}|_{\Gamma_R} = g_R - \alpha \cdot u) &\xrightarrow{1.1,7} \int_{\partial\Omega} v \cdot \mathbf{n}^\top \cdot \mathbf{q} \, ds = \int_{\Gamma_N} g_N \cdot v \, ds + \int_{\Gamma_R} (g_R - \alpha \cdot u) \cdot v \, ds, \\ &\xrightarrow{1.2,4} \int_{\Omega} (\nabla v)^\top \cdot A \cdot \nabla u \, d\mathbf{x} + \int_{\Omega} c \cdot u \cdot v \, d\mathbf{x} + \int_{\Gamma_R} \alpha \cdot u \cdot v \, ds = \int_{\Omega} f \cdot v \, d\mathbf{x} + \int_{\Gamma_N} g_N \cdot v \, ds + \int_{\Gamma_R} g_R \cdot v \, ds. \end{aligned}$$

Both sides are H^1 -continuous in v , so density extends the identity from smooth test functions to V .

$$\begin{aligned} F_u(v) &:= a(u, v) - \ell(v), \quad F_u|_{C^\infty(\overline{\Omega}) \cap V} = 0, \\ v \in V &\xrightarrow{1.2,9} \|v\|_{L^2(\Gamma_N)} + \|v\|_{L^2(\Gamma_R)} \leq C_{\text{tr}} \|v\|_{H^1(\Omega)}, \quad C_u^\Omega := \Lambda_A \|\nabla u\|_{L^2(\Omega)} + \|c \cdot u\|_{L^2(\Omega)} + \|f\|_{L^2(\Omega)}, \\ C_u^\partial &:= C_{\text{tr}} \left(\|\alpha \cdot u\|_{L^2(\Gamma_R)} + \|g_N\|_{L^2(\Gamma_N)} + \|g_R\|_{L^2(\Gamma_R)} \right), \quad C_u := C_u^\Omega + C_u^\partial, \\ |F_u(v)| &\leq C_u \|v\|_{H^1(\Omega)} \implies F_u \in V^*, \quad \overline{C^\infty(\overline{\Omega}) \cap V}^{H^1} = V \xrightarrow{1.2,3} F_u|_V = 0, \\ (F_u|_V = 0, u \in V) &\implies (u \in V, a(u, v) = \ell(v) \ (v \in V)) \xrightarrow{1.2,5} u \in \mathcal{W}_{\text{mix}}. \end{aligned}$$

□

Fact 1.2.9 (Poincaré and trace control). *There are $C_P, C_{\text{tr}} > 0$ such that*

$$\begin{aligned} \|v\|_{L^2(\Omega)} &\leq C_P \|\nabla v\|_{L^2(\Omega)}, \quad \|v\|_{H^1(\Omega)}^2 \leq (1 + C_P^2) \|\nabla v\|_{L^2(\Omega)}^2 \quad (v \in V), \\ \|v\|_{L^2(\Gamma_N)} + \|v\|_{L^2(\Gamma_R)} &\leq C_{\text{tr}} \|v\|_{H^1(\Omega)} \quad (v \in V). \end{aligned}$$

Proof. [4, Section 5.5]. $\square$

Proposition 1.2.10 (Bounds for the mixed form). *Assume $f \in L^2(\Omega)$, $g_N \in L^2(\Gamma_N)$, and $g_R \in L^2(\Gamma_R)$. For $u, v \in V$,*

$$a(v, v) \geq \frac{\lambda_A}{1 + C_P^2} \|v\|_{H^1(\Omega)}^2, \quad |a(u, v)| \leq C_a \|u\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)}, \quad |\ell(v)| \leq C_\ell \|v\|_{H^1(\Omega)},$$

$$C_a = \Lambda_A + \|c\|_{L^\infty(\Omega)} + C_{\text{tr}}^2 \|\alpha\|_{L^\infty(\Gamma_R)}, \quad C_\ell = \|f\|_{L^2(\Omega)} + C_{\text{tr}} \|g_N\|_{L^2(\Gamma_N)} + C_{\text{tr}} \|g_R\|_{L^2(\Gamma_R)}.$$

Proof. Let $u, v \in V$; coercivity uses only ellipticity, since the reaction and Robin terms are nonnegative.

$$c, \alpha \geq 0 \xrightarrow{1.2.4} a(v, v) = \int_{\Omega} (\nabla v)^\top \cdot A \cdot \nabla v \, d\mathbf{x} + \int_{\Omega} c \cdot v^2 \, d\mathbf{x} + \int_{\Gamma_R} \alpha \cdot v^2 \, ds \xrightarrow{1.1.4} a(v, v) \geq \lambda_A \|\nabla v\|_{L^2(\Omega)}^2,$$

$$a(v, v) \geq \lambda_A \|\nabla v\|_{L^2(\Omega)}^2 \xrightarrow{1.2.9} a(v, v) \geq \frac{\lambda_A}{1 + C_P^2} \|v\|_{H^1(\Omega)}^2.$$

For continuity, split a into its diffusion, reaction, and Robin terms and bound each by Cauchy–Schwarz.

$$I_A(u, v) := \int_{\Omega} (\nabla v)^\top \cdot A \cdot \nabla u \, d\mathbf{x}, \quad \xrightarrow{1.1.4} |I_A(u, v)| \leq \Lambda_A \|u\|_V \|v\|_V,$$

$$I_c(u, v) := \int_{\Omega} c \cdot u \cdot v \, d\mathbf{x}, \quad |I_c(u, v)| \leq \|c\|_{L^\infty(\Omega)} \|u\|_V \|v\|_V, \quad I_R(u, v) := \int_{\Gamma_R} \alpha \cdot u \cdot v \, ds,$$

$$|I_R(u, v)| \leq \|\alpha\|_{L^\infty(\Gamma_R)} \|u\|_{L^2(\Gamma_R)} \|v\|_{L^2(\Gamma_R)} \xrightarrow{1.2.9} |I_R(u, v)| \leq C_{\text{tr}}^2 \|\alpha\|_{L^\infty(\Gamma_R)} \|u\|_V \|v\|_V,$$

$$a(u, v) = I_A(u, v) + I_c(u, v) + I_R(u, v) \xrightarrow{1.2.4} |a(u, v)| \leq C_a \|u\|_V \|v\|_V = C_a \|u\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)},$$

$$|\ell(v)| \leq \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} + \|g_N\|_{L^2(\Gamma_N)} \|v\|_{L^2(\Gamma_N)} + \|g_R\|_{L^2(\Gamma_R)} \|v\|_{L^2(\Gamma_R)} \xrightarrow{1.2.9} |\ell(v)| \leq C_\ell \|v\|_{H^1(\Omega)}.$$

$\square$

Fact 1.2.11 (Lax–Milgram). *Let X be a real Hilbert space, $b : X \times X \rightarrow \mathbb{R}$ bounded and coercive, and $L \in X^*$. Then*

$$\exists! w \in X \quad b(w, z) = L(z) \quad (z \in X).$$

Proof. [4, Section 6.2]. $\square$

Theorem 1.2.12 (Existence, uniqueness, and stability). *Assume $g_D = 0$, the hypotheses of Definition 1.1.7, $f \in L^2(\Omega)$, $g_N \in L^2(\Gamma_N)$, and $g_R \in L^2(\Gamma_R)$. Then there is a unique weak mixed solution $u \in V$ in the sense of Definition 1.2.5, and*

$$\|u\|_{H^1(\Omega)} \leq \frac{1 + C_P^2}{\lambda_A} \left(\|f\|_{L^2(\Omega)} + C_{\text{tr}} \|g_N\|_{L^2(\Gamma_N)} + C_{\text{tr}} \|g_R\|_{L^2(\Gamma_R)} \right).$$

Proof. Let V carry $\|\cdot\|_{H^1(\Omega)}$; the form bounds make a coercive and bounded and ℓ continuous, which is exactly the Lax–Milgram frame.

$$\mathcal{D}_D := \{\varphi \in C^\infty(\overline{\Omega}) : \varphi|_{\Gamma_D} = 0\} \xrightarrow{1.2.3} \left(V = \overline{\mathcal{D}_D}^{H^1}, \|\cdot\|_V = \|\cdot\|_{H^1(\Omega)} \right) \implies (V, \|\cdot\|_V) \in \text{Hilb}_{\mathbb{R}},$$

$$v \in V \xrightarrow{1.2.10} a(v, v) \geq \frac{\lambda_A}{1 + C_P^2} \|v\|_{H^1(\Omega)}^2 = \alpha_a \|v\|_V^2, \quad \alpha_a := \frac{\lambda_A}{1 + C_P^2} \xrightarrow{1.1.4} \alpha_a > 0,$$

$$u, v \in V \xrightarrow{1.2.10} |a(u, v)| \leq C_a \|u\|_V \|v\|_V, \quad |\ell(v)| \leq C_\ell \|v\|_V \implies \ell \in V^*,$$

$$((V, \|\cdot\|_V) \in \text{Hilb}_{\mathbb{R}}, a(v, v) \geq \alpha_a \|v\|_V^2, |a(u, v)| \leq C_a \|u\|_V \|v\|_V, \ell \in V^*)$$

$$\xrightarrow{1.2.11} \exists! u \in V : a(u, v) = \ell(v) \quad (v \in V) \xleftrightarrow{1.2.5} \exists! u \in \mathcal{W}_{\text{mix}}.$$

Testing with $v = u$ turns coercivity into stability.

$$\begin{aligned} v = u &\implies a(u, u) = \ell(u) \xrightarrow{1.2.10} \frac{\lambda_A}{1 + C_P^2} \|u\|_{H^1(\Omega)}^2 \leq a(u, u) = \ell(u) \leq C_\ell \|u\|_{H^1(\Omega)}, \\ \frac{\lambda_A}{1 + C_P^2} \|u\|_{H^1(\Omega)}^2 &\leq C_\ell \|u\|_{H^1(\Omega)} \implies \|u\|_{H^1(\Omega)} \leq \frac{1 + C_P^2}{\lambda_A} C_\ell, \\ \|u\|_{H^1(\Omega)} &\leq \frac{1 + C_P^2}{\lambda_A} C_\ell \xrightarrow{1.2.10} \|u\|_{H^1(\Omega)} \leq \frac{1 + C_P^2}{\lambda_A} \left(\|f\|_{L^2(\Omega)} + C_{\text{tr}} \|g_N\|_{L^2(\Gamma_N)} + C_{\text{tr}} \|g_R\|_{L^2(\Gamma_R)} \right). \end{aligned}$$

□

Proposition 1.2.13 (Energy minimization). *Under the hypotheses of Theorem 1.2.12, define*

$$J(v) = \frac{1}{2} a(v, v) - \ell(v) \quad (v \in V).$$

Then u is the weak mixed solution if and only if $J(u) = \min_{v \in V} J(v)$; the minimizer is unique.

Proof. Let $w = v - u \in V$; the form a is symmetric, so J is a parabola along every line through u and its slope at u is the weak residual.

$$\begin{aligned} A = A^T &\xrightarrow{1.2.4} a(u + w, u + w) = a(u, u) + 2a(u, w) + a(w, w) \implies J(u + w) - J(u) = a(u, w) - \ell(w) + \frac{1}{2} a(w, w), \\ u \in \mathcal{W}_{\text{mix}} &\xleftrightarrow{1.2.5} (u \in V, a(u, w) = \ell(w) \ (w \in V)) \implies J(v) - J(u) = \frac{1}{2} a(w, w), \\ J(v) - J(u) &= \frac{1}{2} a(w, w) \xrightarrow{1.2.10} J(v) - J(u) \geq \frac{\lambda_A}{2(1 + C_P^2)} \|w\|_{H^1(\Omega)}^2 \geq 0, \\ J(v) &= J(u) \xrightarrow{1.2.10} \|v - u\|_{H^1(\Omega)} = 0 \implies v = u, \\ J(u) &= \min_{v \in V} J(v) \implies 0 = \left. \frac{d}{dt} \right|_{t=0} J(u + tw) = a(u, w) - \ell(w) \quad (w \in V), \\ (u \in V, a(u, w) &= \ell(w) \ (w \in V)) \xleftrightarrow{1.2.5} u \in \mathcal{W}_{\text{mix}}. \end{aligned}$$

□

1.3. Convection–diffusion.

Definition 1.3.1 (Convection–diffusion form). Let $\varepsilon > 0$, $\beta \in W^{1,\infty}(\Omega; \mathbb{R}^d)$, $c \in L^\infty(\Omega)$, and $f \in L^2(\Omega)$. The strong and weak problems are [1, Section 2.6]

$$\begin{cases} -\varepsilon \Delta u + \beta^\top \cdot \nabla u + c \cdot u = f, & \mathbf{x} \in \Omega, \\ u = 0, & \mathbf{x} \in \partial\Omega, \end{cases}$$

$$a_{\text{cd}}(u, v) := \varepsilon \int_{\Omega} (\nabla v)^\top \cdot \nabla u d\mathbf{x} + \int_{\Omega} v \cdot \beta^\top \cdot \nabla u d\mathbf{x} + \int_{\Omega} c \cdot u \cdot v d\mathbf{x} \quad (u, v \in H_0^1(\Omega)),$$

$$\ell_{\text{cd}}(v) := \int_{\Omega} f \cdot v d\mathbf{x} \quad (v \in H_0^1(\Omega)), \quad u \in H_0^1(\Omega), \quad a_{\text{cd}}(u, v) = \ell_{\text{cd}}(v) \quad (v \in H_0^1(\Omega)).$$

Proposition 1.3.2 (Coercive convection criterion). *If $c - \frac{1}{2} \nabla \cdot \beta \geq 0$ a.e. in Ω , then*

$$a_{\text{cd}}(v, v) \geq \varepsilon \|\nabla v\|_{L^2(\Omega)}^2 \geq \frac{\varepsilon}{1 + C_P^2} \|v\|_{H^1(\Omega)}^2 \quad (v \in H_0^1(\Omega)).$$

Proof. Let $v \in C_c^\infty(\Omega)$; the convection term is half a total divergence plus a $\nabla \cdot \beta$ correction, and compact support kills the total divergence.

$$\begin{aligned} \nabla(v^2) &= 2v \cdot \nabla v, \quad \xrightarrow{1.1.2} \nabla \cdot (\beta v^2) = \beta^\top \cdot \nabla(v^2) + (\nabla \cdot \beta) v^2, \quad \beta v^2 \in W_0^{1,1}(\Omega; \mathbb{R}^d) \xrightarrow{1.2.1} \int_\Omega \nabla \cdot (\beta v^2) d\mathbf{x} = 0, \\ \int_\Omega v \cdot \beta^\top \cdot \nabla v d\mathbf{x} &= \frac{1}{2} \int_\Omega \beta^\top \cdot \nabla(v^2) d\mathbf{x} = \frac{1}{2} \int_\Omega \nabla \cdot (\beta v^2) d\mathbf{x} - \frac{1}{2} \int_\Omega (\nabla \cdot \beta) v^2 d\mathbf{x} = -\frac{1}{2} \int_\Omega (\nabla \cdot \beta) v^2 d\mathbf{x}, \\ v \in C_c^\infty(\Omega) &\xrightarrow{1.3.1} a_{\text{cd}}(v, v) = \varepsilon \|\nabla v\|_{L^2(\Omega)}^2 + \int_\Omega \left(c - \frac{1}{2} \nabla \cdot \beta \right) v^2 d\mathbf{x}, \quad c - \frac{1}{2} \nabla \cdot \beta \geq 0 \implies a_{\text{cd}}(v, v) \geq \varepsilon \|\nabla v\|_{L^2(\Omega)}^2. \end{aligned}$$

Continuity of a_{cd} and density of $C_c^\infty(\Omega)$ in $H_0^1(\Omega)$ extend the inequality to the closure.

$$\begin{aligned} |a_{\text{cd}}(u, v)| &\leq (\varepsilon + \|\beta\|_{L^\infty(\Omega)} + \|c\|_{L^\infty(\Omega)}) \|u\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)} \xrightarrow{1.3.1} a_{\text{cd}} \in \mathcal{B}(H_0^1(\Omega) \times H_0^1(\Omega), \mathbb{R}), \\ R(v) &:= a_{\text{cd}}(v, v) - \varepsilon \|\nabla v\|_{L^2(\Omega)}^2, \quad a_{\text{cd}} \in \mathcal{B}(H_0^1(\Omega) \times H_0^1(\Omega), \mathbb{R}) \implies R \in C(H_0^1(\Omega); \mathbb{R}), \\ \left(R \in C(H_0^1(\Omega); \mathbb{R}), R|_{C_c^\infty(\Omega)} \geq 0, \overline{C_c^\infty(\Omega)}^{H^1} &= H_0^1(\Omega) \right) \implies R|_{H_0^1(\Omega)} \geq 0, \\ R|_{H_0^1(\Omega)} \geq 0 &\implies a_{\text{cd}}(v, v) \geq \varepsilon \|\nabla v\|_{L^2(\Omega)}^2 \xrightarrow{1.2.9} a_{\text{cd}}(v, v) \geq \frac{\varepsilon}{1 + C_p^2} \|v\|_{H^1(\Omega)}^2 \quad (v \in H_0^1(\Omega)). \end{aligned}$$

□

2. GALERKIN FINITE ELEMENTS

Conforming restriction, elementwise assembly, and stationary correction convert the weak problem of Definition 1.2.5 into finite-dimensional algebra.

2.1. Matrix form.

Definition 2.1.1 (Conforming simplicial P_1 space). Let $\mathcal{T}_h$ be a conforming simplicial triangulation of $\overline{\Omega}$ resolving the $\Gamma_D, \Gamma_N, \Gamma_R$ of Definition 1.1.6, and let V be the space of Definition 1.2.3 [1, Sections 3.2 and 4.1].

$$h = \max_{T \in \mathcal{T}_h} \text{diam}(T), \quad V_h = \{v_h \in C^0(\overline{\Omega}) \mid v_h|_T \in \mathcal{P}_1(T) \ (T \in \mathcal{T}_h), v_h|_{\Gamma_D} = 0\} \subset V.$$

For the free vertices $\mathbf{z}_1, \dots, \mathbf{z}_N$, define

$$\psi_j(\mathbf{z}_i) = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases} \quad V_h = \text{span}\{\psi_1, \dots, \psi_N\}, \quad I_h v = \sum_{j=1}^N v(\mathbf{z}_j) \psi_j.$$

Proposition 2.1.2 (Discrete Galerkin well-posedness). *Under the hypotheses of Proposition 1.2.10, there is a unique $u_h \in V_h$ satisfying*

$$a(u_h, v_h) = \ell(v_h) \quad (v_h \in V_h).$$

Proof. Let $\alpha_h = \lambda_A / (1 + C_p^2)$, $M_h = C_a$, and $C_h = C_\ell$; conformity hands the form bounds down to V_h , and a coercive problem on a finite-dimensional space is a Lax–Milgram problem.

$$v_h, w_h \in V_h \xrightarrow{2.1.1} v_h, w_h \in V \xrightarrow{1.2.10} \begin{cases} |a(v_h, w_h)| \leq M_h \|v_h\|_V \|w_h\|_V, \\ a(v_h, v_h) \geq \alpha_h \|v_h\|_V^2, \\ |\ell(v_h)| \leq C_h \|v_h\|_V, \end{cases} \quad \alpha_h > 0,$$

$$(V_h \subset V, \dim V_h = N < \infty, M_h < \infty, \alpha_h > 0, C_h < \infty)$$

$$\xrightarrow{1.2.11} \exists! u_h \in V_h : a(u_h, v_h) = \ell(v_h) \quad (v_h \in V_h).$$

□

Definition 2.1.3 (Galerkin matrix system). Let $u_h = \sum_{j=1}^N U_j \psi_j$ be the solution of Proposition 2.1.2; define $\mathbf{U}$, K , and $\mathbf{F}$ by [1, Sections 4.1–4.6]

$$\mathbf{U} = \begin{bmatrix} U_1 \\ U_2 \\ \vdots \\ U_N \end{bmatrix}, \quad K_{ij} = a(\psi_j, \psi_i), \quad F_i = \ell(\psi_i), \quad K = [K_{ij}]_{i,j=1}^N, \quad \mathbf{F} = \begin{bmatrix} F_1 \\ F_2 \\ \vdots \\ F_N \end{bmatrix}.$$

$$u_h = \sum_{j=1}^N U_j \psi_j \xrightarrow{1.2.4} a(u_h, \psi_i) = \sum_{j=1}^N a(\psi_j, \psi_i) \cdot U_j \quad (1 \leq i \leq N),$$

$$(a(u_h, \psi_i) = \ell(\psi_i))_{i=1}^N \iff \begin{bmatrix} a(\psi_1, \psi_1) & \cdots & a(\psi_N, \psi_1) \\ \vdots & \ddots & \vdots \\ a(\psi_1, \psi_N) & \cdots & a(\psi_N, \psi_N) \end{bmatrix} \cdot \begin{bmatrix} U_1 \\ \vdots \\ U_N \end{bmatrix} = \begin{bmatrix} \ell(\psi_1) \\ \vdots \\ \ell(\psi_N) \end{bmatrix},$$

$$(K_{ij} = a(\psi_j, \psi_i), F_i = \ell(\psi_i)) \implies K \cdot \mathbf{U} = \mathbf{F}.$$

Definition 2.1.4 (Assembled mixed system). For the form and load of Definition 1.2.4, the entries of Definition 2.1.3 expand to

$$K_{ij} = \int_{\Omega} (\nabla \psi_i)^T \cdot A \cdot \nabla \psi_j d\mathbf{x} + \int_{\Omega} c \cdot \psi_j \cdot \psi_i d\mathbf{x} + \int_{\Gamma_R} \alpha \cdot \psi_j \cdot \psi_i ds,$$

$$F_i = \int_{\Omega} f \cdot \psi_i d\mathbf{x} + \int_{\Gamma_N} g_N \cdot \psi_i ds + \int_{\Gamma_R} g_R \cdot \psi_i ds,$$

$$(K \cdot \mathbf{U})_i = \sum_{j=1}^N \left(\int_{\Omega} (\nabla \psi_i)^T \cdot A \cdot \nabla \psi_j d\mathbf{x} + \int_{\Omega} c \cdot \psi_j \cdot \psi_i d\mathbf{x} + \int_{\Gamma_R} \alpha \cdot \psi_j \cdot \psi_i ds \right) \cdot U_j = F_i.$$

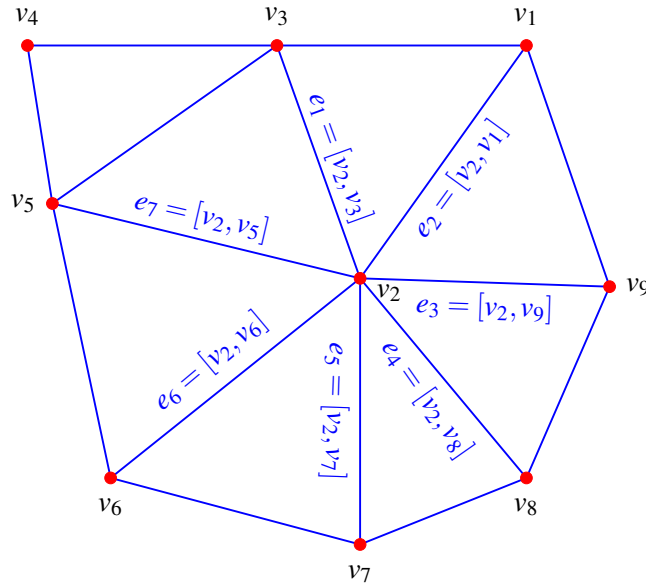


FIGURE 2. A triangulated planar graph with central vertex v_2 .

Proposition 2.1.5 (Galerkin orthogonality and Céa estimate). *Assume $V_h \subset V$, $a(u, v) = \ell(v)$ for $v \in V$, $a(u_h, v_h) = \ell(v_h)$ for $v_h \in V_h$, and*

$$|a(v, w)| \leq M_a \|v\|_V \|w\|_V, \quad a(v, v) \geq \alpha_a \|v\|_V^2, \quad M_a \geq \alpha_a > 0.$$

Then [2, Chapter 4]

$$a(u - u_h, w_h) = 0 \quad (w_h \in V_h), \quad \|u - u_h\|_V \leq \frac{M_a}{\alpha_a} \inf_{v_h \in V_h} \|u - v_h\|_V.$$

Proof. Fix $w_h, v_h \in V_h$ and set $e_h = u - u_h$; the error is a -orthogonal to V_h , so coercivity against continuity compares it with any competitor.

$$\begin{aligned} w_h \in V_h \subset V &\xrightarrow{1.2.5, 2.1.2} a(u, w_h) = \ell(w_h) = a(u_h, w_h) \implies a(e_h, w_h) = 0, \\ v_h - u_h \in V_h &\implies a(e_h, v_h - u_h) = 0, \quad a(e_h, e_h) = a(e_h, u - v_h) + a(e_h, v_h - u_h) = a(e_h, u - v_h), \\ \alpha_a \|e_h\|_V^2 &\leq a(e_h, e_h) = a(e_h, u - v_h) \leq M_a \|e_h\|_V \|u - v_h\|_V, \\ e_h = 0 \quad \vee \quad &\left(e_h \neq 0 \implies \|e_h\|_V \leq \frac{M_a}{\alpha_a} \|u - v_h\|_V \right) \implies \|u - u_h\|_V \leq \frac{M_a}{\alpha_a} \inf_{v_h \in V_h} \|u - v_h\|_V. \end{aligned}$$

□

Corollary 2.1.6 (Positive-definite Galerkin matrix). *If a is symmetric and coercive, then the matrix K of Definition 2.1.3 is symmetric positive definite [2, Chapter 4].*

Proof. Choose $\alpha_a > 0$ such that $a(v, v) \geq \alpha_a \|v\|_V^2$, $\mathbf{c} = [c_1 \ \cdots \ c_N]^\top \in \mathbb{R}^N \setminus \{\mathbf{0}\}$, and $v_h = \sum_{j=1}^N c_j \psi_j$; the quadratic form of K is a evaluated on the spanned function.

$$\begin{aligned} \mathbf{c} \neq \mathbf{0} &\xrightarrow{2.1.1} v_h \neq 0, \\ \mathbf{c}^\top \cdot K \cdot \mathbf{c} &\xrightarrow{2.1.3} \mathbf{c}^\top \cdot K \cdot \mathbf{c} = [c_1 \ \cdots \ c_N] \cdot \begin{bmatrix} a(\psi_1, \psi_1) & \cdots & a(\psi_N, \psi_1) \\ \vdots & \ddots & \vdots \\ a(\psi_1, \psi_N) & \cdots & a(\psi_N, \psi_N) \end{bmatrix} \cdot \begin{bmatrix} c_1 \\ \vdots \\ c_N \end{bmatrix}, \\ \mathbf{c}^\top \cdot K \cdot \mathbf{c} &= \sum_{i=1}^N \sum_{j=1}^N c_i \cdot a(\psi_j, \psi_i) \cdot c_j = a\left(\sum_{j=1}^N c_j \psi_j, \sum_{i=1}^N c_i \psi_i\right) = a(v_h, v_h) \geq \alpha_a \|v_h\|_V^2 > 0, \\ a(v, w) = a(w, v) &\implies K_{ij} = a(\psi_j, \psi_i) = a(\psi_i, \psi_j) = K_{ji} \implies K = K^\top, \quad \mathbf{c}^\top \cdot K \cdot \mathbf{c} > 0 \quad (\mathbf{c} \neq \mathbf{0}). \end{aligned}$$

□

2.2. Dimensional assembly.

Example 2.2.1 (The nonuniform one-dimensional stiffness system). Let

$$0 = x_0 < x_1 < \cdots < x_n < x_{n+1} = 1, \quad h_i = x_i - x_{i-1} \quad (1 \leq i \leq n+1),$$

and derive the nodal system for [1, Sections 4.1 and 5.3]

$$\begin{cases} -u''(x) = f(x), & 0 < x < 1, \\ u(0) = 0, \\ u(1) = 0, \end{cases} \quad \xleftrightarrow{1.1.3} (d, f, g) = (1, f, 0).$$

For $1 \leq i \leq n$,

$$\psi_i(x) = \begin{cases} \frac{x - x_{i-1}}{h_i}, & x_{i-1} \leq x \leq x_i, \\ \frac{x_{i+1} - x}{h_{i+1}}, & x_i \leq x \leq x_{i+1}, \\ 0, & x \notin [x_{i-1}, x_{i+1}], \end{cases} \quad \psi'_i(x) = \begin{cases} \frac{1}{h_i}, & x_{i-1} < x < x_i, \\ -\frac{1}{h_{i+1}}, & x_i < x < x_{i+1}, \\ 0, & x \notin (x_{i-1}, x_{i+1}). \end{cases}$$

$$\begin{aligned}
K_{ii} &\stackrel{2.1.4}{\implies} K_{ii} = \int_0^1 \psi'_i(x) \psi'_i(x) dx = \int_{x_{i-1}}^{x_i} \frac{1}{h_i^2} dx + \int_{x_i}^{x_{i+1}} \frac{1}{h_{i+1}^2} dx = \frac{1}{h_i} + \frac{1}{h_{i+1}}, \\
K_{i,i-1} &= K_{i-1,i} = \int_{x_{i-1}}^{x_i} \left(-\frac{1}{h_i}\right) \cdot \frac{1}{h_i} dx = -\frac{x_i - x_{i-1}}{h_i^2} = -\frac{1}{h_i}, \\
K_{i,i+1} &= K_{i+1,i} = \int_{x_i}^{x_{i+1}} \frac{1}{h_{i+1}} \cdot \left(-\frac{1}{h_{i+1}}\right) dx = -\frac{x_{i+1} - x_i}{h_{i+1}^2} = -\frac{1}{h_{i+1}}, \\
|i-j| > 1 &\implies \text{supp}(\psi_i) \cap \text{supp}(\psi_j) = \emptyset \implies K_{ij} = 0, \\
K &= \begin{bmatrix} \frac{1}{h_1} + \frac{1}{h_2} & -\frac{1}{h_2} & 0 & \cdots & 0 \\ -\frac{1}{h_2} & \frac{1}{h_2} + \frac{1}{h_3} & -\frac{1}{h_3} & \ddots & \vdots \\ 0 & -\frac{1}{h_3} & \frac{1}{h_3} + \frac{1}{h_4} & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -\frac{1}{h_n} \\ 0 & \cdots & 0 & -\frac{1}{h_n} & \frac{1}{h_n} + \frac{1}{h_{n+1}} \end{bmatrix}. \\
F_i &= \int_0^1 f(x) \psi_i(x) dx = \frac{1}{h_i} \int_{x_{i-1}}^{x_i} f(x)(x - x_{i-1}) dx + \frac{1}{h_{i+1}} \int_{x_i}^{x_{i+1}} f(x)(x_{i+1} - x) dx, \\
F_i &\approx \frac{h_i}{2} f\left(\frac{x_{i-1} + x_i}{2}\right) + \frac{h_{i+1}}{2} f\left(\frac{x_i + x_{i+1}}{2}\right), \\
f = 1 &\implies F_i = \frac{1}{h_i} \left[\frac{(x - x_{i-1})^2}{2}\right]_{x_{i-1}}^{x_i} - \frac{1}{h_{i+1}} \left[\frac{(x_{i+1} - x)^2}{2}\right]_{x_i}^{x_{i+1}} = \frac{h_i}{2} + \frac{h_{i+1}}{2}, \\
\mathbf{U} &= [U_1 \quad \cdots \quad U_n]^\top, \quad \mathbf{F} = [F_1 \quad \cdots \quad F_n]^\top, \quad K \cdot \mathbf{U} = \mathbf{F} \stackrel{2.1.3}{\implies} u_h = \sum_{i=1}^n U_i \psi_i.
\end{aligned}$$

Example 2.2.2 (Four-element hand assembly). Let $f = 1$, $x_j = j/4$, $h = 1/4$, and $E_j = [x_j, x_{j+1}]$.

$$\begin{aligned}
D_j &= \begin{bmatrix} \psi'_{j,L} \\ \psi'_{j,R} \end{bmatrix} = \begin{bmatrix} -1/h \\ 1/h \end{bmatrix} = \begin{bmatrix} -4 \\ 4 \end{bmatrix}, \\
K^{(j)} &= \int_{x_j}^{x_{j+1}} D_j \cdot D_j^\top dx = \int_{x_j}^{x_{j+1}} \begin{bmatrix} -4 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} -4 & 4 \end{bmatrix} dx = \frac{1}{4} \begin{bmatrix} (-4)(-4) & (-4)(4) \\ (4)(-4) & (4)(4) \end{bmatrix} = \begin{bmatrix} 4 & -4 \\ -4 & 4 \end{bmatrix}, \\
\mathbf{F}^{(j)} &= \int_{x_j}^{x_{j+1}} \begin{bmatrix} \psi_{j,L}(x) \\ \psi_{j,R}(x) \end{bmatrix} dx = \begin{bmatrix} \int_0^h (1-t/h) dt \\ \int_0^h t/h dt \end{bmatrix} = \begin{bmatrix} h-h/2 \\ h/2 \end{bmatrix} = \begin{bmatrix} 1/8 \\ 1/8 \end{bmatrix}, \\
K^{\text{full}} &= \begin{bmatrix} 4 & -4 & 0 & 0 & 0 \\ -4 & 4 & -4 & 0 & 0 \\ 0 & -4 & 4 & -4 & 0 \\ 0 & 0 & -4 & 4 & -4 \\ 0 & 0 & 0 & -4 & 4 \end{bmatrix} = \begin{bmatrix} 4 & -4 & 0 & 0 & 0 \\ -4 & 8 & -4 & 0 & 0 \\ 0 & -4 & 8 & -4 & 0 \\ 0 & 0 & -4 & 8 & -4 \\ 0 & 0 & 0 & -4 & 4 \end{bmatrix}, \\
\mathbf{F}^{\text{full}} &= \begin{bmatrix} 1/8 \\ 1/8 + 1/8 \\ 1/8 + 1/8 \\ 1/8 + 1/8 \\ 1/8 \end{bmatrix} = \begin{bmatrix} 1/8 \\ 1/4 \\ 1/4 \\ 1/4 \\ 1/8 \end{bmatrix},
\end{aligned}$$

$$\begin{aligned}
U_0 = U_4 = 0 &\xrightarrow{2.2.1} \begin{bmatrix} 8 & -4 & 0 \\ -4 & 8 & -4 \\ 0 & -4 & 8 \end{bmatrix} \cdot \begin{bmatrix} U_1 \\ U_2 \\ U_3 \end{bmatrix} = \begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}, \\
\begin{bmatrix} 8 & -4 & 0 \\ -4 & 8 & -4 \\ 0 & -4 & 8 \end{bmatrix} \cdot \begin{bmatrix} 3/32 \\ 1/8 \\ 3/32 \end{bmatrix} &= \begin{bmatrix} 8 \cdot \frac{3}{32} - 4 \cdot \frac{1}{8} \\ -4 \cdot \frac{3}{32} + 8 \cdot \frac{1}{8} - 4 \cdot \frac{3}{32} \\ -4 \cdot \frac{1}{8} + 8 \cdot \frac{3}{32} \end{bmatrix} = \begin{bmatrix} 3/4 - 1/2 \\ -3/8 + 1 - 3/8 \\ -1/2 + 3/4 \end{bmatrix} = \begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}, \\
(a(v, w) = a(w, v), a(v, v) \geq \alpha_a \|v\|_V^2, \alpha_a > 0) &\xrightarrow{2.1.6} K \succ 0 \implies \ker(K) = \{\mathbf{0}\}, \\
\left(K \cdot \begin{bmatrix} 3/32 \\ 1/8 \\ 3/32 \end{bmatrix} = \mathbf{F}, \ker(K) = \{\mathbf{0}\} \right) &\implies \mathbf{U} = \begin{bmatrix} 3/32 \\ 1/8 \\ 3/32 \end{bmatrix}, \quad u_h = \frac{3}{32}\psi_1 + \frac{1}{8}\psi_2 + \frac{3}{32}\psi_3, \\
-u'' = 1 &\xrightarrow{1.1.11} u(x) = -\frac{x^2}{2} + C_1x + C_0, \quad u(0) = u(1) = 0 \implies (C_0, C_1) = \left(0, \frac{1}{2}\right), \\
u(x) = \frac{x(1-x)}{2}, \quad \begin{bmatrix} u(1/4) \\ u(1/2) \\ u(3/4) \end{bmatrix} &= \begin{bmatrix} 3/32 \\ 1/8 \\ 3/32 \end{bmatrix} \xrightarrow{2.1.1} u_h = I_h u, \\
x = x_j + t, \quad 0 < t < h &\implies u(x) - u_h(x) = \frac{1}{2}t(h-t) > 0 \implies u_h \neq u \quad \text{on } (x_j, x_{j+1}).
\end{aligned}$$

Example 2.2.3 (Exact interpolation errors). Let $N \in \{2, 3, \dots\}$, $h = N^{-1}$, $x_j = jh$, $E_j = [x_j, x_{j+1}]$, and $t = x - x_j \in [0, h]$.

$$\begin{aligned}
u(x) &= \frac{x(1-x)}{2}, \quad u_j = u(x_j) = \frac{jh - j^2h^2}{2}, \quad u_0 = u_N = 0, \quad \mathbf{u} = [u_1 \quad \dots \quad u_{N-1}]^\top, \\
2u_j - u_{j-1} - u_{j+1} &= jh - j^2h^2 - \frac{(j-1)h - (j-1)^2h^2}{2} - \frac{(j+1)h - (j+1)^2h^2}{2} = h^2, \\
h_i &= h, \quad f = 1 \xrightarrow{2.2.1} (K \cdot \mathbf{u})_j = \frac{-u_{j-1} + 2u_j - u_{j+1}}{h} = h = F_j \quad (1 \leq j \leq N-1), \\
K \cdot \mathbf{u} &= \mathbf{F}, \quad (a(v, w) = a(w, v), a(v, v) \geq \alpha_a \|v\|_V^2, \alpha_a > 0) \xrightarrow{2.1.6} K \succ 0 \implies \ker(K) = \{\mathbf{0}\}, \\
K \cdot \mathbf{u} &= \mathbf{F} = K \cdot \mathbf{U}, \quad \ker(K) = \{\mathbf{0}\} \implies \mathbf{U} = \mathbf{u} \xrightarrow{2.1.1} u_h = I_h u, \\
I_h u(x_j + t) &= \left(1 - \frac{t}{h}\right)u(x_j) + \frac{t}{h}u(x_{j+1}), \quad e(t) = u(x_j + t) - I_h u(x_j + t) = \frac{1}{2}t(h-t), \quad e'(t) = \frac{h}{2} - t, \\
\int_0^h |e'(t)|^2 dt &= \int_0^h \left(\frac{h}{2} - t\right)^2 dt = \int_0^h \left(t^2 - ht + \frac{h^2}{4}\right) dt = \left[\frac{t^3}{3} - \frac{ht^2}{2} + \frac{h^2t}{4}\right]_0^h = \frac{h^3}{12}, \\
\int_0^h |e(t)|^2 dt &= \frac{1}{4} \int_0^h t^2(h-t)^2 dt = \frac{1}{4} \int_0^h (h^2t^2 - 2ht^3 + t^4) dt = \frac{1}{4} \left[\frac{h^2t^3}{3} - \frac{ht^4}{2} + \frac{t^5}{5}\right]_0^h = \frac{h^5}{120}, \\
\|u - u_h\|_{H^1(0,1)}^2 &= \sum_{j=0}^{N-1} \int_{x_j}^{x_{j+1}} |(u - u_h)'(x)|^2 dx = \sum_{j=0}^{N-1} \frac{h^3}{12} = N \cdot \frac{h^3}{12} = \frac{h^2}{12}, \\
\|u - u_h\|_{L^2(0,1)}^2 &= \sum_{j=0}^{N-1} \int_{x_j}^{x_{j+1}} |u(x) - u_h(x)|^2 dx = \sum_{j=0}^{N-1} \frac{h^5}{120} = N \cdot \frac{h^5}{120} = \frac{h^4}{120}, \\
|u - u_h|_{H^1(0,1)} &= \frac{h}{\sqrt{12}}, \quad \|u - u_h\|_{L^2(0,1)} = \frac{h^2}{\sqrt{120}}, \\
h \mapsto \frac{h}{2} &\implies \frac{h}{\sqrt{12}} \mapsto \frac{1}{2} \frac{h}{\sqrt{12}}, \quad \frac{h^2}{\sqrt{120}} \mapsto \frac{1}{4} \frac{h^2}{\sqrt{120}}.
\end{aligned}$$

Example 2.2.4 (Manufactured reaction–diffusion solution). Let

$$\begin{aligned}
 u(x, y) &= 1 + x + y + x^2 + y^2, \quad (A, c) = (I_2, 1) \xrightarrow{1.1.7} \begin{cases} -\Delta u + u = f, & (x, y) \in \Omega, \\ u = g, & (x, y) \in \partial\Omega. \end{cases} \\
 u(x, y) &= 1 + x + y + x^2 + y^2 \xrightarrow{1.1.1} \nabla u = \begin{bmatrix} \partial_x u \\ \partial_y u \end{bmatrix} = \begin{bmatrix} 1 + 2x \\ 1 + 2y \end{bmatrix}, \quad D^2 u = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \\
 \xrightarrow{1.1.2} \Delta u &= \text{tr}(D^2 u) = 2 + 2 = 4, \quad f = -\Delta u + u = x^2 + y^2 + x + y - 3, \quad g = (1 + x + y + x^2 + y^2)|_{\partial\Omega}, \\
 \mathbf{v}_0 &= (0, 0), \quad \mathbf{v}_1 = (1, 0), \quad \mathbf{v}_2 = (0, 1), \quad \begin{bmatrix} u(\mathbf{v}_0) \\ u(\mathbf{v}_1) \\ u(\mathbf{v}_2) \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix}, \\
 [\lambda_0 \quad \lambda_1 \quad \lambda_2] \cdot \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix} &= 1 \cdot \lambda_0 + 3\lambda_1 + 3\lambda_2, \quad (\lambda_0, \lambda_1, \lambda_2) = (1 - x - y, x, y) \xrightarrow{2.1.1} I_h u = 1 + 2x + 2y.
 \end{aligned}$$

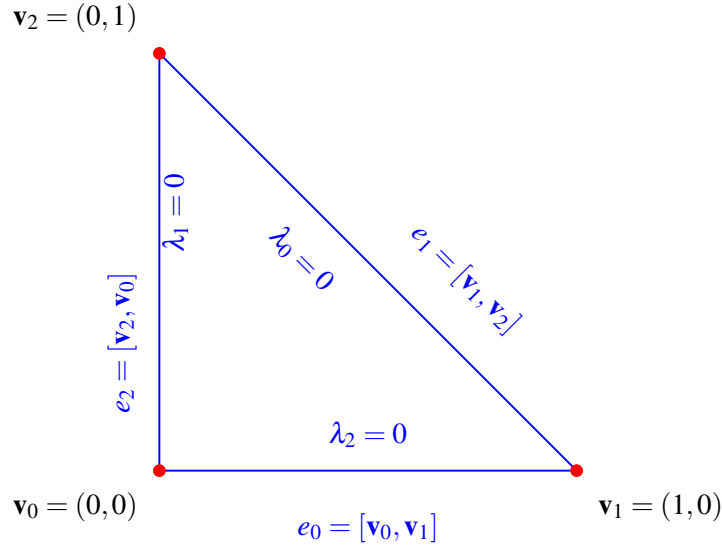


FIGURE 3. Reference triangle and barycentric zero sets.

Example 2.2.5 (Reference-triangle element matrices). On the cell of Figure 3 [1, Sections 3.3 and 4.2],

$$\begin{aligned}
 \hat{T} &= \{(x, y) \in \mathbb{R}^2 : x \geq 0, y \geq 0, x + y \leq 1\}, \quad |\hat{T}| = \frac{1}{2}, \quad \lambda_0 + \lambda_1 + \lambda_2 = 1, \\
 (\lambda_0, \lambda_1, \lambda_2) &= (1 - x - y, x, y) \xrightarrow{1.1.1} \nabla \lambda_0 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \quad \nabla \lambda_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \nabla \lambda_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad D = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}, \\
 K_{\hat{T}} &= \int_{\hat{T}} D^T \cdot D \, dx \, dy = |\hat{T}| D^T \cdot D = \frac{1}{2} \begin{bmatrix} -1 & -1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}, \\
 K_{\hat{T}} &= \frac{1}{2} \begin{bmatrix} (-1)(-1) + (-1)(-1) & (-1)(1) + (-1)(0) & (-1)(0) + (-1)(1) \\ (1)(-1) + (0)(-1) & (1)(1) + (0)(0) & (1)(0) + (0)(1) \\ (0)(-1) + (1)(-1) & (0)(1) + (1)(0) & (0)(0) + (1)(1) \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix},
 \end{aligned}$$

$$\begin{aligned}
\int_{\hat{T}} \lambda_0^2 dx dy &= \int_0^1 \int_0^{1-x} (1-x-y)^2 dy dx = \int_0^1 \frac{(1-x)^3}{3} dx = \frac{1}{12}, \\
\int_{\hat{T}} \lambda_1^2 dx dy &= \int_0^1 \int_0^{1-x} x^2 dy dx = \int_0^1 x^2(1-x) dx = \frac{1}{12}, \\
\int_{\hat{T}} \lambda_2^2 dx dy &= \int_0^1 \int_0^{1-x} y^2 dy dx = \int_0^1 \frac{(1-x)^3}{3} dx = \frac{1}{12}, \\
\int_{\hat{T}} \lambda_0 \lambda_1 dx dy &= \int_0^1 \int_0^{1-x} x(1-x-y) dy dx = \frac{1}{2} \int_0^1 x(1-x)^2 dx = \frac{1}{24}, \\
\int_{\hat{T}} \lambda_0 \lambda_2 dx dy &= \int_0^1 \int_0^{1-x} y(1-x-y) dy dx = \frac{1}{6} \int_0^1 (1-x)^3 dx = \frac{1}{24}, \\
\int_{\hat{T}} \lambda_1 \lambda_2 dx dy &= \int_0^1 \int_0^{1-x} xy dy dx = \frac{1}{2} \int_0^1 x(1-x)^2 dx = \frac{1}{24}, \\
M_{\hat{T}} &= \int_{\hat{T}} \begin{bmatrix} \lambda_0 \\ \lambda_1 \\ \lambda_2 \end{bmatrix} \cdot [\lambda_0 \quad \lambda_1 \quad \lambda_2] dx dy = \begin{bmatrix} 1/12 & 1/24 & 1/24 \\ 1/24 & 1/12 & 1/24 \\ 1/24 & 1/24 & 1/12 \end{bmatrix} = \frac{1}{24} \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}, \\
\int_{\hat{T}} \lambda_0 dx dy &= \int_0^1 \int_0^{1-x} (1-x-y) dy dx = \frac{1}{2} \int_0^1 (1-x)^2 dx = \frac{1}{6}, \\
\int_{\hat{T}} \lambda_1 dx dy &= \int_0^1 \int_0^{1-x} x dy dx = \int_0^1 x(1-x) dx = \frac{1}{6}, \quad \int_{\hat{T}} \lambda_2 dx dy = \frac{1}{2} \int_0^1 (1-x)^2 dx = \frac{1}{6}, \\
\mathbf{F}_{\hat{T}} &= \begin{bmatrix} \int_{\hat{T}} \lambda_0 dx dy \\ \int_{\hat{T}} \lambda_1 dx dy \\ \int_{\hat{T}} \lambda_2 dx dy \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad K_{\hat{T}} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2-1-1 \\ -1+1+0 \\ -1+0+1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\} \subseteq \ker(K_{\hat{T}}).
\end{aligned}$$

Proposition 2.2.6 (Affine transport of a triangle matrix). *Let $F_T(\hat{\mathbf{x}}) = B_T \hat{\mathbf{x}} + \mathbf{b}_T$ map $\hat{T}$ onto T , let B_T be nonsingular, let $\psi_i = \hat{\lambda}_i \circ F_T^{-1}$, and let a_T be scalar and constant on T . Then [1, Sections 3.3 and 4.2]*

$$\begin{aligned}
\hat{D} &= \begin{bmatrix} \nabla \hat{\lambda}_0 & \nabla \hat{\lambda}_1 & \nabla \hat{\lambda}_2 \end{bmatrix}, \quad \nabla \psi_i = B_T^{-\top} \cdot \nabla \hat{\lambda}_i, \quad d\mathbf{x} = |\det B_T| d\hat{\mathbf{x}}, \\
K_T &= a_T |\det B_T| |\hat{T}| \hat{D}^\top \cdot B_T^{-1} \cdot B_T^{-\top} \cdot \hat{D}, \quad M_T = |\det B_T| M_{\hat{T}}.
\end{aligned}$$

Proof. Let $\mathbf{x} = F_T(\hat{\mathbf{x}})$; the chain rule transports gradients by $B_T^{-\top}$ and the substitution rule transports volumes by $|\det B_T|$.

$$\begin{aligned}
\hat{\mathbf{x}} &= F_T^{-1}(\mathbf{x}), \quad DF_T^{-1} = B_T^{-1}, \quad \psi_i(\mathbf{x}) = \hat{\lambda}_i(F_T^{-1}(\mathbf{x})) \xrightarrow{1,1,1} \nabla \psi_i(\mathbf{x}) = B_T^{-\top} \cdot \nabla \hat{\lambda}_i(\hat{\mathbf{x}}), \\
(K_T)_{ij} &= \int_T a_T (\nabla \psi_i)^\top \cdot \nabla \psi_j d\mathbf{x} = a_T |\det B_T| \int_{\hat{T}} \left(B_T^{-\top} \cdot \nabla \hat{\lambda}_i \right)^\top \cdot \left(B_T^{-\top} \cdot \nabla \hat{\lambda}_j \right) d\hat{\mathbf{x}}, \\
(K_T)_{ij} &= a_T |\det B_T| \int_{\hat{T}} (\nabla \hat{\lambda}_i)^\top \cdot B_T^{-1} \cdot B_T^{-\top} \cdot \nabla \hat{\lambda}_j d\hat{\mathbf{x}}, \\
\nabla \hat{\lambda}_i, B_T^{-1}, B_T^{-\top} &\text{ constant on } \hat{T} \implies K_T = a_T |\det B_T| |\hat{T}| \hat{D}^\top \cdot B_T^{-1} \cdot B_T^{-\top} \cdot \hat{D}.
\end{aligned}$$

The mass matrix transports by the volume factor alone.

$$(M_T)_{ij} = \int_T \psi_j \psi_i d\mathbf{x} = |\det B_T| \int_{\hat{T}} \hat{\lambda}_j \hat{\lambda}_i d\hat{\mathbf{x}} \xrightarrow{2,2,5} M_T = |\det B_T| M_{\hat{T}}.$$

□

2.3. Stationary iteration.

Definition 2.3.1 (Stationary splitting). Let $K = M - N$, where $M \in \mathbb{R}^{N \times N}$ and $\det M \neq 0$, and define [7, Chapter 3]

$$G = M^{-1} \cdot N, \quad \mathbf{U}^{(k+1)} = G \cdot \mathbf{U}^{(k)} + M^{-1} \cdot \mathbf{F}.$$

Lemma 2.3.2 (Error propagation). For $K \cdot \mathbf{U} = \mathbf{F}$, the iteration of Definition 2.3.1, and $\mathbf{e}^{(k)} = \mathbf{U} - \mathbf{U}^{(k)}$,

$$\mathbf{e}^{(k+1)} = G \cdot \mathbf{e}^{(k)}, \quad \mathbf{e}^{(k)} = G^k \cdot \mathbf{e}^{(0)}.$$

Proof. Let $\mathbf{e}^{(k)} = \mathbf{U} - \mathbf{U}^{(k)}$; the exact solution is a fixed point of the iteration, and subtracting the update kills the constant term.

$$\begin{aligned} K \cdot \mathbf{U} &= \mathbf{F}, \quad K = M - N \xrightarrow{2.3.1} (M - N) \cdot \mathbf{U} = \mathbf{F} \iff M \cdot \mathbf{U} = N \cdot \mathbf{U} + \mathbf{F}, \\ M \cdot \mathbf{U} &= N \cdot \mathbf{U} + \mathbf{F} \iff \mathbf{U} = M^{-1} \cdot N \cdot \mathbf{U} + M^{-1} \cdot \mathbf{F} \xrightarrow{2.3.1} \mathbf{U} = G \cdot \mathbf{U} + M^{-1} \cdot \mathbf{F}, \\ \mathbf{e}^{(k+1)} &= G \cdot \mathbf{U} + M^{-1} \cdot \mathbf{F} - G \cdot \mathbf{U}^{(k)} - M^{-1} \cdot \mathbf{F} = G \cdot (\mathbf{U} - \mathbf{U}^{(k)}) = G \cdot \mathbf{e}^{(k)}, \\ \mathbf{e}^{(k+1)} &= G \cdot \mathbf{e}^{(k)} \implies \mathbf{e}^{(k)} = G^k \cdot \mathbf{e}^{(0)}. \end{aligned}$$

□

Theorem 2.3.3 (Stationary convergence criterion). The iteration of Definition 2.3.1 converges to the solution of $K \cdot \mathbf{U} = \mathbf{F}$ for every $\mathbf{U}^{(0)}$ if and only if $\rho(G) < 1$. For every subordinate matrix norm,

$$\|G\| < 1 \implies \rho(G) < 1 \implies \mathbf{e}^{(k)} \longrightarrow \mathbf{0}.$$

Proof. Let $\mathbf{e}^{(k)} = G^k \cdot \mathbf{e}^{(0)}$ as in Lemma 2.3.2; the powers of G decide everything.

$$\begin{aligned} (\mathbf{U}^{(k)} \rightarrow \mathbf{U} \quad \forall \mathbf{U}^{(0)} \in \mathbb{R}^N) &\xrightarrow{2.3.2} (G^k \cdot \mathbf{e}^{(0)} \rightarrow \mathbf{0} \quad \forall \mathbf{e}^{(0)} \in \mathbb{R}^N), \\ (G^k \cdot \mathbf{e}^{(0)} \rightarrow \mathbf{0} \quad \forall \mathbf{e}^{(0)} \in \mathbb{R}^N) &\iff G^k \rightarrow \mathbf{0} \xrightarrow{[7, \text{Chapter 3}]} \rho(G) < 1, \\ \|G\| < 1 &\implies \|\mathbf{e}^{(k)}\| = \|G^k \cdot \mathbf{e}^{(0)}\| \leq \|G^k\| \cdot \|\mathbf{e}^{(0)}\| \leq \|G\|^k \cdot \|\mathbf{e}^{(0)}\| \longrightarrow 0. \end{aligned}$$

□

Example 2.3.4 (Jacobi iteration for the four-element matrix). For Jacobi, let $D = \text{diag}(K)$, $M = D$, and $N = D - K$.

$$\begin{aligned} K &\xrightarrow{2.2.2} K = \begin{bmatrix} 8 & -4 & 0 \\ -4 & 8 & -4 \\ 0 & -4 & 8 \end{bmatrix}, \quad D = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 8 \end{bmatrix}, \quad N = D - K = \begin{bmatrix} 0 & 4 & 0 \\ 4 & 0 & 4 \\ 0 & 4 & 0 \end{bmatrix}, \\ G &\xrightarrow{2.3.1} G = D^{-1} \cdot N = \begin{bmatrix} 1/8 & 0 & 0 \\ 0 & 1/8 & 0 \\ 0 & 0 & 1/8 \end{bmatrix} \cdot \begin{bmatrix} 0 & 4 & 0 \\ 4 & 0 & 4 \\ 0 & 4 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1/2 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1/2 & 0 \end{bmatrix}, \\ \det(G - \lambda I_3) &= \det \begin{bmatrix} -\lambda & 1/2 & 0 \\ 1/2 & -\lambda & 1/2 \\ 0 & 1/2 & -\lambda \end{bmatrix} = (-\lambda) \det \begin{bmatrix} -\lambda & 1/2 \\ 1/2 & -\lambda \end{bmatrix} - \frac{1}{2} \det \begin{bmatrix} 1/2 & 1/2 \\ 0 & -\lambda \end{bmatrix}, \\ \det(G - \lambda I_3) &= (-\lambda) \left(\lambda^2 - \frac{1}{4} \right) - \frac{1}{2} \left(-\frac{\lambda}{2} \right) = -\lambda^3 + \frac{\lambda}{2} = -\lambda \left(\lambda^2 - \frac{1}{2} \right), \\ \text{spec}(G) &= \left\{ -\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right\}, \quad \rho(G) = \frac{1}{\sqrt{2}} < 1 \xrightarrow{2.3.3} \mathbf{U}^{(k)} \longrightarrow \mathbf{U} \quad (\mathbf{U}^{(0)} \in \mathbb{R}^3). \end{aligned}$$

Proposition 2.3.5 (Two-grid error propagator). *Let $K_h \cdot \mathbf{U} = \mathbf{F}$, let $K_h \in \mathbb{R}^{N_h \times N_h}$, $K_H \in \mathbb{R}^{N_H \times N_H}$ be nonsingular, $R \in \mathbb{R}^{N_H \times N_h}$, $P \in \mathbb{R}^{N_h \times N_H}$, and let $S \in \mathbb{R}^{N_h \times N_h}$ be the smoothing error propagator. One two-grid cycle with v pre-smoothing steps satisfies*

$$E_{\text{TG}} = (I_{N_h} - P \cdot K_H^{-1} \cdot R \cdot K_h) \cdot S^v = (K_h^{-1} - P \cdot K_H^{-1} \cdot R) \cdot K_h \cdot S^v,$$

$$\|E_{\text{TG}}\| \leq \|K_h^{-1} - P \cdot K_H^{-1} \cdot R\| \cdot \|K_h \cdot S^v\|.$$

Proof. Let $\mathbf{e}^{(k,v)} = S^v \cdot \mathbf{e}^{(k)}$; smoothing damps the error, the coarse solve corrects the smoothed residual, and the propagator is their product.

$$\begin{aligned} \mathbf{e}^{(k)} &= \mathbf{U} - \mathbf{U}^{(k)} \xrightarrow{2.3.2} \mathbf{e}^{(k,v)} = S^v \cdot \mathbf{e}^{(k)}, \\ \mathbf{r}^{(k,v)} &= \mathbf{F} - K_h \cdot \mathbf{U}^{(k,v)} = K_h \cdot \mathbf{U} - K_h \cdot \mathbf{U}^{(k,v)} = K_h \cdot (\mathbf{U} - \mathbf{U}^{(k,v)}) = K_h \cdot \mathbf{e}^{(k,v)}, \\ \mathbf{d}_H &= K_H^{-1} \cdot R \cdot \mathbf{r}^{(k,v)} = K_H^{-1} \cdot R \cdot K_h \cdot \mathbf{e}^{(k,v)}. \end{aligned}$$

The correction prolongates the coarse defect and subtracts it, and the propagator factors through the residual.

$$\begin{aligned} \mathbf{U}^{(k+1)} &= \mathbf{U}^{(k,v)} + P \cdot \mathbf{d}_H \implies \mathbf{e}^{(k+1)} = \mathbf{U} - \mathbf{U}^{(k,v)} - P \cdot \mathbf{d}_H, \\ \mathbf{e}^{(k+1)} &= \mathbf{e}^{(k,v)} - P \cdot K_H^{-1} \cdot R \cdot K_h \cdot \mathbf{e}^{(k,v)} = (I_{N_h} - P \cdot K_H^{-1} \cdot R \cdot K_h) \cdot S^v \cdot \mathbf{e}^{(k)}, \\ I_{N_h} - P \cdot K_H^{-1} \cdot R \cdot K_h &= K_h^{-1} \cdot K_h - P \cdot K_H^{-1} \cdot R \cdot K_h = (K_h^{-1} - P \cdot K_H^{-1} \cdot R) \cdot K_h, \\ E_{\text{TG}} &= (K_h^{-1} - P \cdot K_H^{-1} \cdot R) \cdot K_h \cdot S^v \implies \|E_{\text{TG}}\| \leq \|K_h^{-1} - P \cdot K_H^{-1} \cdot R\| \cdot \|K_h \cdot S^v\|. \end{aligned}$$

□

3. ORIENTED SIMPLICIAL COMPLEXES

Throughout, K is a finite oriented simplicial complex with $|K| = \overline{\Omega} \subset \mathbb{R}^d$, and all chain and cochain spaces are real.

3.1. Oriented simplicial meshes.

Definition 3.1.1 (Simplex and conforming simplicial complex). Let $\mathbf{v}_0, \dots, \mathbf{v}_p \in \mathbb{R}^d$ be affinely independent, and write $\tau \preceq \sigma$ when τ is a face of σ [8, Chapter 1].

$$\begin{aligned} [\mathbf{v}_0, \dots, \mathbf{v}_p] &= \left\{ [\mathbf{v}_0 \ \cdots \ \mathbf{v}_p] \cdot \begin{bmatrix} t_0 \\ \vdots \\ t_p \end{bmatrix} \mid t_i \geq 0 \ (0 \leq i \leq p), \quad [1 \ \cdots \ 1] \cdot \begin{bmatrix} t_0 \\ \vdots \\ t_p \end{bmatrix} = 1 \right\}, \\ \sigma \in K, \quad \tau \preceq \sigma &\implies \tau \in K, \quad \sigma, \tau \in K \implies \sigma \cap \tau \in \{\emptyset\} \cup \{\eta : \eta \preceq \sigma, \eta \preceq \tau\}, \\ |K| &:= \bigcup_{\sigma \in K} \sigma, \quad |K| = \overline{\Omega}. \end{aligned}$$

Example 3.1.2 (Reference triangle in matrix form). Let $\mathbf{v}_0 = [0, 0]^\top$, $\mathbf{v}_1 = [1, 0]^\top$, and $\mathbf{v}_2 = [0, 1]^\top$.

$$\begin{aligned} \xrightarrow{3.1.1} [\mathbf{v}_0 \ \mathbf{v}_1 \ \mathbf{v}_2] \cdot \begin{bmatrix} t_0 \\ \mathbf{t}_1 \\ \mathbf{t}_2 \end{bmatrix} &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} t_0 \\ \mathbf{t}_1 \\ \mathbf{t}_2 \end{bmatrix} = \begin{bmatrix} 0 \cdot t_0 + 1 \cdot \mathbf{t}_1 + 0 \cdot \mathbf{t}_2 \\ 0 \cdot t_0 + 0 \cdot \mathbf{t}_1 + 1 \cdot \mathbf{t}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{t}_1 \\ \mathbf{t}_2 \end{bmatrix}, \\ t_0 + t_1 + t_2 &= 1, \quad t_0, t_1, t_2 \geq 0 \iff t_1, t_2 \geq 0, \quad t_1 + t_2 \leq 1, \\ [\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2] &= \left\{ \begin{bmatrix} \mathbf{t}_1 \\ \mathbf{t}_2 \end{bmatrix} \mid t_1 \geq 0, \quad t_2 \geq 0, \quad t_1 + t_2 \leq 1 \right\}. \end{aligned}$$

Definition 3.1.3 (Orientation, chains, and boundary). For $\sigma = [\mathbf{v}_0, \dots, \mathbf{v}_p]$, an even vertex permutation preserves σ and an odd vertex permutation replaces σ by $-\sigma$ [8, Chapter 1].

$$K_p := \{\sigma \in K : \dim \sigma = p\}, \quad C_p(K) := \text{span}\{\sigma : \sigma \in K_p\} = \left\{ \sum_{\sigma \in K_p} a_\sigma \cdot \sigma \mid a_\sigma \in \mathbb{R} \right\},$$

$$\begin{aligned} \partial_0 &:= 0, \quad \partial_p[\mathbf{v}_0, \dots, \mathbf{v}_p] := \sum_{i=0}^p (-1)^i [\mathbf{v}_0, \dots, \widehat{\mathbf{v}}_i, \dots, \mathbf{v}_p] \quad (p \geq 1), \\ c &= \sum_{\sigma \in K_p} a_\sigma \cdot \sigma \implies \partial_p c = \sum_{\sigma \in K_p} a_\sigma \cdot \partial_p \sigma, \quad \partial_p : C_p(K) \longrightarrow C_{p-1}(K). \end{aligned}$$

Example 3.1.4 (Tetrahedral and linear-chain boundaries). Let $\sigma = [\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3]$ and $c = \pi[\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2] - \eta[\mathbf{v}_2, \mathbf{v}_1, \mathbf{v}_3]$.

$$\begin{aligned} \sigma &\xrightarrow{3.1.3} \partial_3 \sigma = [\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3] - [\mathbf{v}_0, \mathbf{v}_2, \mathbf{v}_3] + [\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_3] - [\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2], \\ \partial_2 c &\xrightarrow{3.1.3} \partial_2 c = \pi \cdot ([\mathbf{v}_1, \mathbf{v}_2] - [\mathbf{v}_0, \mathbf{v}_2] + [\mathbf{v}_0, \mathbf{v}_1]) - \eta \cdot ([\mathbf{v}_1, \mathbf{v}_3] - [\mathbf{v}_2, \mathbf{v}_3] + [\mathbf{v}_2, \mathbf{v}_1]). \end{aligned}$$

In the edge order $[\mathbf{v}_0, \mathbf{v}_1], [\mathbf{v}_0, \mathbf{v}_2], [\mathbf{v}_0, \mathbf{v}_3], [\mathbf{v}_1, \mathbf{v}_2], [\mathbf{v}_1, \mathbf{v}_3], [\mathbf{v}_2, \mathbf{v}_3]$,

$$\begin{bmatrix} 1 & 0 \\ -1 & 0 \\ 0 & 0 \\ 1 & -1 \\ 0 & 1 \\ 0 & -1 \end{bmatrix} \cdot \begin{bmatrix} \pi \\ -\eta \end{bmatrix} = \begin{bmatrix} 1 \cdot \pi + 0 \cdot (-\eta) \\ (-1) \cdot \pi + 0 \cdot (-\eta) \\ 0 \cdot \pi + 0 \cdot (-\eta) \\ 1 \cdot \pi + (-1) \cdot (-\eta) \\ 0 \cdot \pi + 1 \cdot (-\eta) \\ 0 \cdot \pi + (-1) \cdot (-\eta) \end{bmatrix} = \begin{bmatrix} \pi \\ -\pi \\ 0 \\ \pi + \eta \\ -\eta \\ \eta \end{bmatrix}.$$

Theorem 3.1.5 (The boundary of a boundary vanishes). For every $p \geq 1$,

$$\partial_{p-1} \circ \partial_p = 0.$$

Proof. Let $\sigma = [\mathbf{v}_0, \dots, \mathbf{v}_p]$; deleting $\mathbf{v}_i$ then $\mathbf{v}_j$ and deleting $\mathbf{v}_j$ then $\mathbf{v}_i$ produce the same face with opposite signs.

$$\begin{aligned} p = 1 &\xrightarrow{3.1.3} \partial_0 \partial_1 = 0, \\ p \geq 2 &\xrightarrow{3.1.3} \partial_{p-1} \partial_p \sigma = \sum_{i=0}^p \sum_{\substack{0 \leq j \leq p \\ j \neq i}} \varepsilon_{ij} [\mathbf{v}_0, \dots, \widehat{\mathbf{v}}_i, \dots, \widehat{\mathbf{v}}_j, \dots, \mathbf{v}_p], \\ 0 \leq i < j \leq p &\implies \varepsilon_{ij} = (-1)^i \cdot (-1)^{j-1} = (-1)^{i+j-1}, \quad \varepsilon_{ji} = (-1)^j \cdot (-1)^i = (-1)^{i+j}, \\ \varepsilon_{ij} + \varepsilon_{ji} &= (-1)^{i+j-1} + (-1)^{i+j} = (-1)^{i+j-1} \cdot (1 - 1) = 0, \\ \partial_{p-1} \partial_p \sigma &= \sum_{0 \leq i < j \leq p} (\varepsilon_{ij} + \varepsilon_{ji}) [\mathbf{v}_0, \dots, \widehat{\mathbf{v}}_i, \dots, \widehat{\mathbf{v}}_j, \dots, \mathbf{v}_p] = 0, \\ c &= \sum_{\sigma \in K_p} a_\sigma \cdot \sigma \xrightarrow{3.1.3} \partial_{p-1} \partial_p c = \sum_{\sigma \in K_p} a_\sigma \cdot \partial_{p-1} \partial_p \sigma = 0. \end{aligned}$$

□

Definition 3.1.6 (Chain complex and incidence matrices). Choose ordered oriented bases $K_p = \{\sigma_1^p, \dots, \sigma_{N_p}^p\}$.

$$\begin{aligned} C_{-1}(K) &= C_{d+1}(K) = \{0\}, \quad \partial_0 = \partial_{d+1} = 0, \\ 0 &\longrightarrow C_d(K) \xrightarrow{\partial_d} C_{d-1}(K) \xrightarrow{\partial_{d-1}} \dots \xrightarrow{\partial_2} C_1(K) \xrightarrow{\partial_1} C_0(K) \longrightarrow 0, \\ \partial_p \sigma_j^p &= \sum_{i=1}^{N_{p-1}} b_{ij}^{(p)} \cdot \sigma_i^{p-1}, \quad b_{ij}^{(p)} =: [\sigma_j^p : \sigma_i^{p-1}] \in \{-1, 0, 1\} \quad (1 \leq p \leq d), \end{aligned}$$

$$B_p := \begin{bmatrix} b_{11}^{(p)} & \cdots & b_{1N_p}^{(p)} \\ \vdots & \ddots & \vdots \\ b_{N_{p-1}1}^{(p)} & \cdots & b_{N_{p-1}N_p}^{(p)} \end{bmatrix} \in \mathbb{R}^{N_{p-1} \times N_p},$$

$$c = \sum_{j=1}^{N_p} a_j \cdot \sigma_j^p, \quad [c]_{K_p} = \begin{bmatrix} a_1 \\ \vdots \\ a_{N_p} \end{bmatrix}, \quad [\partial_p c]_{K_{p-1}} = B_p \cdot [c]_{K_p} = \begin{bmatrix} \sum_{j=1}^{N_p} b_{1j}^{(p)} a_j \\ \vdots \\ \sum_{j=1}^{N_p} b_{N_{p-1}j}^{(p)} a_j \end{bmatrix}.$$

Proposition 3.1.7 (Boundaries are cycles). *For the indicated degrees,*

$$\text{ran}(\partial_{p+1}) \subseteq \ker(\partial_p) \quad (0 \leq p < d), \quad B_p \cdot B_{p+1} = 0 \quad (1 \leq p < d).$$

Proof. Let $z = \partial_{p+1}c$; the square of the boundary vanishes termwise, and the matrices inherit the identity from the operators.

$$z \in \text{ran}(\partial_{p+1}) \implies \exists c \in C_{p+1}(K) : z = \partial_{p+1}c \xrightarrow{3.1.5} \partial_p z = \partial_p \partial_{p+1}c = 0 \implies z \in \ker(\partial_p),$$

$$1 \leq p < d, \quad \partial_p \partial_{p+1} = 0 \xrightarrow{3.1.6} B_p \cdot B_{p+1} = 0.$$

□

Definition 3.1.8 (Shape regularity and the Delaunay condition). Let $\{K_h\}_h$ triangulate $\bar{\Omega}$, set $\mathcal{T}_h := (K_h)_d$, $\mathcal{V}_h := (K_h)_0$, and for $T \in \mathcal{T}_h$ let $h_T = \text{diam}(T)$, ρ_T be its inradius, and D_T its closed circumdisk [9, Chapter 2]; the family is shape regular, and in two dimensions Delaunay, under the respective conditions

$$\exists \kappa \geq 1 : \sup_h \max_{T \in \mathcal{T}_h} \frac{h_T}{\rho_T} \leq \kappa, \quad d = 2, \quad \forall T \in \mathcal{T}_h : \mathring{D}_T \cap (\mathcal{V}_h \setminus T) = \emptyset.$$

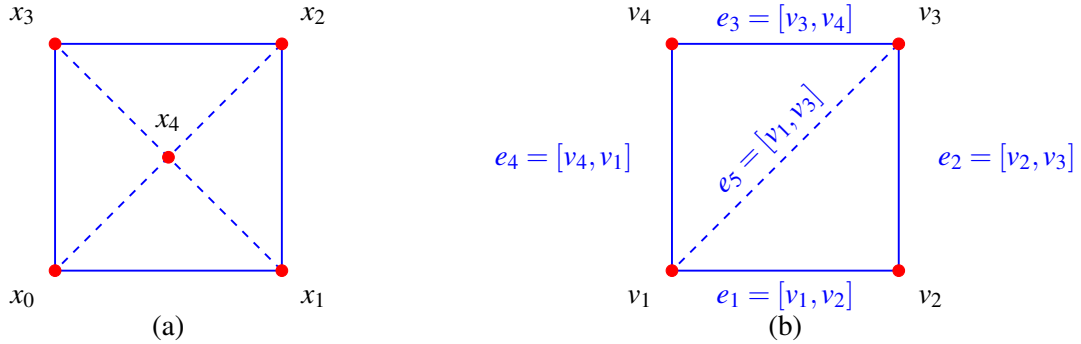


FIGURE 4. Four-triangle and two-triangle oriented square meshes.

Example 3.1.9 (The four-triangle chain count). For Figure 4(a), orient the boundary counterclockwise and every radial edge toward x_4 .

$$e_0 = [x_0, x_1], \quad e_1 = [x_1, x_2], \quad e_2 = [x_2, x_3], \quad e_3 = [x_3, x_0],$$

$$e_4 = [x_0, x_4], \quad e_5 = [x_1, x_4], \quad e_6 = [x_2, x_4], \quad e_7 = [x_3, x_4],$$

$$C_0(K) = \text{span}\{x_0, x_1, x_2, x_3, x_4\}, \quad C_1(K) = \text{span}\{e_0, e_1, e_2, e_3, e_4, e_5, e_6, e_7\},$$

$$C_2(K) = \text{span}\{f_0, f_1, f_2, f_3\} \xrightarrow{3.1.3} (\dim C_0(K), \dim C_1(K), \dim C_2(K)) = (5, 8, 4),$$

$$\chi(K) \xrightarrow{3.1.11} \chi(K) = 5 - 8 + 4 = 1.$$

Example 3.1.10 (Incidence matrices and $\partial^2 = 0$). For Figure 4(b), use the displayed vertex, edge, and face orientations.

$$\partial_1 e_1 = v_2 - v_1, \quad \partial_1 e_2 = v_3 - v_2, \quad \partial_1 e_3 = v_4 - v_3, \quad \partial_1 e_4 = v_1 - v_4, \quad \partial_1 e_5 = v_3 - v_1,$$

$$(\partial_1 e_1, \partial_1 e_2, \partial_1 e_3, \partial_1 e_4, \partial_1 e_5) \xrightarrow{3.1.6} B_1 = \begin{bmatrix} -1 & 0 & 0 & 1 & -1 \\ 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 1 & -1 & 0 \end{bmatrix},$$

$$\partial_2 T_1 = e_1 + e_2 - e_5, \quad \partial_2 T_2 = e_3 + e_4 + e_5, \quad (\partial_2 T_1, \partial_2 T_2) \xrightarrow{3.1.6} B_2 = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix},$$

$$B_1 \cdot B_2 = \begin{bmatrix} -1 & 0 & 0 & 1 & -1 \\ 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 1 & -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix},$$

$$B_1 \cdot B_2 = \begin{bmatrix} (-1) \cdot 1 + 0 \cdot 1 + 0 \cdot 0 + 1 \cdot 0 + (-1) \cdot (-1) & (-1) \cdot 0 + 0 \cdot 0 + 0 \cdot 1 + 1 \cdot 1 + (-1) \cdot 1 \\ 1 \cdot 1 + (-1) \cdot 1 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot (-1) & 1 \cdot 0 + (-1) \cdot 0 + 0 \cdot 1 + 0 \cdot 1 + 0 \cdot 1 \\ 0 \cdot 1 + 1 \cdot 1 + (-1) \cdot 0 + 0 \cdot 0 + 1 \cdot (-1) & 0 \cdot 0 + 1 \cdot 0 + (-1) \cdot 1 + 0 \cdot 1 + 1 \cdot 1 \\ 0 \cdot 1 + 0 \cdot 1 + 1 \cdot 0 + (-1) \cdot 0 + 0 \cdot (-1) & 0 \cdot 0 + 0 \cdot 0 + 1 \cdot 1 + (-1) \cdot 1 + 0 \cdot 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix},$$

$$T_1, T_2 \in C_2(K) \xrightarrow{3.1.5} \partial_1 \partial_2 T_1 = \partial_1 \partial_2 T_2 = 0 \xrightarrow{3.1.6} B_1 \cdot B_2 = 0.$$

Definition 3.1.11 (Simplicial homology and Euler characteristic). For $0 \leq p \leq d$, set

$$Z_p(K) := \ker(\partial_p), \quad \mathcal{B}_p(K) := \text{ran}(\partial_{p+1}), \quad \mathcal{B}_p(K) \xrightarrow{3.1.7} \mathcal{B}_p(K) \subseteq Z_p(K),$$

$$H_p(K) := \frac{Z_p(K)}{\mathcal{B}_p(K)}, \quad \beta_p(K) := \dim H_p(K), \quad \chi(K) := \sum_{p=0}^d (-1)^p \dim C_p(K).$$

Example 3.1.12 (Cycles, boundaries, and Betti numbers). For the two-triangle square of Example 3.1.10, write $c = \sum_{j=1}^5 a_j \cdot e_j$ and solve $B_1 \cdot \mathbf{a} = 0$.

$$\partial_1 c \xrightarrow{3.1.3} \partial_1 c = (-a_1 + a_4 - a_5)v_1 + (a_1 - a_2)v_2 + (a_2 - a_3 + a_5)v_3 + (a_3 - a_4)v_4,$$

$$B_1 \cdot \mathbf{a} = \begin{bmatrix} -1 & 0 & 0 & 1 & -1 \\ 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 1 & -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{bmatrix} = \begin{bmatrix} -a_1 + a_4 - a_5 \\ a_1 - a_2 \\ a_2 - a_3 + a_5 \\ a_3 - a_4 \end{bmatrix} = \mathbf{0},$$

$$a_2 = a_1, \quad a_4 = a_3, \quad a_5 = a_3 - a_1, \quad \textcolor{red}{s} := a_1, \quad \textcolor{blue}{t} := a_3,$$

$$c = \textcolor{red}{s} \cdot (e_1 + e_2 - e_5) + \textcolor{blue}{t} \cdot (e_3 + e_4 + e_5) = \textcolor{red}{s} \cdot \partial_2 T_1 + \textcolor{blue}{t} \cdot \partial_2 T_2,$$

$$\ker(B_1) = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\} = \text{ran}(B_2) \xrightarrow{3.1.11} H_1(K) = \{0\},$$

$$\det \begin{bmatrix} -1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix} = -1 \neq 0, \quad \det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 1 \neq 0, \quad \text{rank}(B_1) = 3, \quad \text{rank}(B_2) = 2,$$

$$\beta_0 = \dim C_0(K) - \text{rank}(B_1) = 4 - 3 = 1, \quad \beta_1 = \dim \ker(B_1) - \text{rank}(B_2) = (5 - 3) - 2 = 0,$$

$$\beta_2 = \dim \ker(B_2) = 2 - 2 = 0 \xrightarrow{3.1.11} \chi(K) = 4 - 5 + 2 = 1 = \beta_0 - \beta_1 + \beta_2.$$

3.2. Cochains.

Definition 3.2.1 (Cochains and dual bases). Let $K_p = \{\sigma_1^p, \dots, \sigma_{N_p}^p\}$ be ordered [8, Chapter 5], and set $K_p^* := \{(\sigma_1^p)^*, \dots, (\sigma_{N_p}^p)^*\}$.

$$C^p(K) := C_p(K)^* := \{\ell : C_p(K) \rightarrow \mathbb{R} \mid \ell(a \cdot c + b \cdot c') = a \cdot \ell(c) + b \cdot \ell(c') \quad (a, b \in \mathbb{R}, c, c' \in C_p(K))\},$$

$$(\sigma_i^p)^*(\sigma_j^p) = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases} \quad c = \sum_{j=1}^{N_p} a_j \cdot \sigma_j^p, \quad \ell = \sum_{i=1}^{N_p} \ell_i \cdot (\sigma_i^p)^*,$$

$$\ell(c) = \sum_{i=1}^{N_p} \sum_{j=1}^{N_p} \ell_i a_j \cdot (\sigma_i^p)^*(\sigma_j^p) = \sum_{i=1}^{N_p} \ell_i a_i = [\ell_1 \quad \dots \quad \ell_{N_p}] \cdot \begin{bmatrix} a_1 \\ \vdots \\ a_{N_p} \end{bmatrix}.$$

Definition 3.2.2 (Coboundary). For $c \in C^p(K)$, define $\delta^p c \in C^{p+1}(K)$ by

$$(\delta^p c)(\tau) := c(\partial_{p+1} \tau) \quad (\tau \in C_{p+1}(K)), \quad \delta^p : C^p(K) \longrightarrow C^{p+1}(K),$$

$$C^{-1}(K) = C^{d+1}(K) = \{0\}, \quad \delta^{-1} = \delta^d = 0.$$

Proposition 3.2.3 (Coboundaries square to zero). For $0 \leq p < d$,

$$\delta^{p+1} \circ \delta^p = 0, \quad [\delta^p] = B_{p+1}^\top.$$

Proof. Let $c \in C^p(K)$ and $\tau \in C_{p+2}(K)$; the coboundary is the adjoint of the boundary, so squares transpose to squares.

$$c \in C^p(K), \quad \tau \in C_{p+2}(K) \xrightarrow{3.2.2} (\delta^{p+1} \delta^p c)(\tau) = (\delta^p c)(\partial_{p+2} \tau) = c(\partial_{p+1} \partial_{p+2} \tau),$$

$$\partial_{p+1} \partial_{p+2} = 0 \xrightarrow{3.1.5} (\delta^{p+1} \delta^p c)(\tau) = c(0) = 0 \implies \delta^{p+1} \delta^p = 0,$$

$$(\delta^p (\sigma_i^p)^*)(\sigma_j^{p+1}) \xrightarrow{3.2.2} (\sigma_i^p)^*(\partial_{p+1} \sigma_j^{p+1}) = (\sigma_i^p)^* \left(\sum_{k=1}^{N_p} b_{kj}^{(p+1)} \cdot \sigma_k^p \right) \xrightarrow{3.1.6} b_{ij}^{(p+1)},$$

$$[\delta^p]_{ji} = b_{ij}^{(p+1)} = (B_{p+1}^\top)_{ji} \implies [\delta^p] = B_{p+1}^\top.$$

□

Example 3.2.4 (Coboundary matrices for the two-triangle square). The incidence matrices of Example 3.1.10 give

$$[\delta^0] \xrightarrow{3.2.3} B_1^\top = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \\ 1 & 0 & 0 & -1 \\ -1 & 0 & 1 & 0 \end{bmatrix}, \quad [\delta^1] \xrightarrow{3.2.3} B_2^\top = \begin{bmatrix} 1 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix},$$

$$[\delta^1] \cdot [\delta^0] = \begin{bmatrix} 1 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \\ 1 & 0 & 0 & -1 \\ -1 & 0 & 1 & 0 \end{bmatrix}.$$

The first two and last two columns of the product are

$$\begin{bmatrix} 1 \cdot (-1) + 1 \cdot 0 + 0 \cdot 0 + 0 \cdot 1 + (-1) \cdot (-1) & 1 \cdot 1 + 1 \cdot (-1) + 0 \cdot 0 + 0 \cdot 0 + (-1) \cdot 0 \\ 0 \cdot (-1) + 0 \cdot 0 + 1 \cdot 0 + 1 \cdot 1 + 1 \cdot (-1) & 0 \cdot 1 + 0 \cdot (-1) + 1 \cdot 0 + 1 \cdot 0 + 1 \cdot 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix},$$

$$\begin{bmatrix} 1 \cdot 0 + 1 \cdot 1 + 0 \cdot (-1) + 0 \cdot 0 + (-1) \cdot 1 & 1 \cdot 0 + 1 \cdot 0 + 0 \cdot 1 + 0 \cdot (-1) + (-1) \cdot 0 \\ 0 \cdot 0 + 0 \cdot 1 + 1 \cdot (-1) + 1 \cdot 0 + 1 \cdot 1 & 0 \cdot 0 + 0 \cdot 0 + 1 \cdot 1 + 1 \cdot (-1) + 1 \cdot 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix},$$

$$[\delta^1] \cdot [\delta^0] = B_2^\top \cdot B_1^\top = (B_1 \cdot B_2)^\top \xrightarrow{3.1.10} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Definition 3.2.5 (Simplicial cohomology). For $0 \leq p \leq d$, set

$$Z^p(K) := \ker(\delta^p), \quad \mathcal{B}^p(K) := \text{ran}(\delta^{p-1}), \quad \mathcal{B}^0(K) = \text{ran}(\delta^{-1}) \xrightarrow{3.2.2} \mathcal{B}^0(K) = \{0\} \subseteq Z^0(K),$$

$$b \in \mathcal{B}^p(K) \implies \exists c \in C^{p-1}(K) : b = \delta^{p-1}c \xrightarrow{3.2.3} \delta^p b = \delta^p \delta^{p-1}c = 0 \implies b \in Z^p(K),$$

$$H^p(K) := \frac{Z^p(K)}{\mathcal{B}^p(K)}.$$

Example 3.2.6 (The coboundary of a vertex cochain). In Figure 2, order the remaining edges by

$$e_8 = [v_5, v_3], \quad e_9 = [v_4, v_3], \quad e_{10} = [v_3, v_1], \quad e_{11} = [v_1, v_9], \quad e_{12} = [v_9, v_8],$$

$$e_{13} = [v_8, v_7], \quad e_{14} = [v_7, v_6], \quad e_{15} = [v_6, v_5], \quad e_{16} = [v_5, v_4].$$

The rows of $B_1^\top$ indexed by $e_1, \dots, e_7$, in the vertex order $(v_1, \dots, v_9)$, are

$$S = \begin{bmatrix} 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad S \cdot [v_2^*] = S \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \cdot 0 + (-1) \cdot 1 + 1 \cdot 0 \\ 1 \cdot 0 + (-1) \cdot 1 + 0 \cdot 0 \\ 0 \cdot 0 + (-1) \cdot 1 + 1 \cdot 0 \\ 0 \cdot 0 + (-1) \cdot 1 + 1 \cdot 0 \\ 0 \cdot 0 + (-1) \cdot 1 + 1 \cdot 0 \\ 0 \cdot 0 + (-1) \cdot 1 + 1 \cdot 0 \\ 0 \cdot 0 + (-1) \cdot 1 + 1 \cdot 0 \\ 0 \cdot 0 + (-1) \cdot 1 + 1 \cdot 0 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ -1 \\ -1 \\ -1 \\ -1 \\ -1 \\ -1 \end{bmatrix},$$

$$e_j \cap \{v_2\} = \emptyset \quad (8 \leq j \leq 16) \xrightarrow{3.2.2} (\delta^0 v_2^*)(e_j) = v_2^*(\partial_1 e_j) = 0,$$

$$[\delta^0 v_2^*]_{(e_1^*, \dots, e_{16}^*)} \xrightarrow{3.2.3} B_1^\top \cdot [v_2^*] = \begin{bmatrix} -1 \\ -1 \\ -1 \\ -1 \\ -1 \\ -1 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \delta^0 v_2^* = - \sum_{j=1}^7 e_j^*, \quad \sum_{j=1}^{16} (\delta^0 v_2^*)(e_j) = -7.$$

Definition 3.2.7 (Differential forms and de Rham cohomology). Let $\Omega^p(\overline{\Omega})$ be the smooth real p -forms on a neighborhood of $\overline{\Omega}$ [13, Section 2] [14, Sections 4.1–4.2].

$$\Omega^0(\overline{\Omega}) \xrightarrow{d} \Omega^1(\overline{\Omega}) \xrightarrow{d} \dots \xrightarrow{d} \Omega^d(\overline{\Omega}), \quad d \circ d = 0, \quad \Omega^{-1}(\overline{\Omega}) = \{0\},$$

$$H_{\text{dR}}^p(\Omega) := \frac{\ker(d|_{\Omega^p(\overline{\Omega})})}{\text{ran}(d|_{\Omega^{p-1}(\overline{\Omega})})},$$

$$C^\infty(\overline{\Omega}) \xrightarrow{\nabla} C^\infty(\overline{\Omega}; \mathbb{R}^3) \xrightarrow{\nabla \times} C^\infty(\overline{\Omega}; \mathbb{R}^3) \xrightarrow{\nabla \cdot} C^\infty(\overline{\Omega}).$$

Proposition 3.2.8 (Coordinate form of the vector-proxy operators). Let $u \in C^2(\overline{\Omega})$ and $\mathbf{F} \in C^2(\overline{\Omega}; \mathbb{R}^3)$.

$$\nabla u = \begin{bmatrix} \partial_1 u \\ \partial_2 u \\ \partial_3 u \end{bmatrix}, \quad \nabla \times \mathbf{F} = \begin{bmatrix} \partial_2 F_3 - \partial_3 F_2 \\ \partial_3 F_1 - \partial_1 F_3 \\ \partial_1 F_2 - \partial_2 F_1 \end{bmatrix}, \quad \nabla \cdot \mathbf{F} = \partial_1 F_1 + \partial_2 F_2 + \partial_3 F_3,$$

$$\nabla \times (\nabla u) = \mathbf{0}, \quad \nabla \cdot (\nabla \times \mathbf{F}) = 0.$$

Proof. Let $u \in C^2(\overline{\Omega})$ and $\mathbf{F} \in C^2(\overline{\Omega}; \mathbb{R}^3)$; equality of mixed partials [4, Appendix C] cancels every term in pairs.

$$\begin{aligned} \xrightarrow{1.1.1, 1.1.2} \nabla u &= \begin{bmatrix} \partial_1 u \\ \partial_2 u \\ \partial_3 u \end{bmatrix}, \quad \nabla \cdot \mathbf{F} = [\partial_1 \quad \partial_2 \quad \partial_3] \cdot \begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix} = \partial_1 F_1 + \partial_2 F_2 + \partial_3 F_3, \\ \nabla \times \mathbf{F} &= \begin{bmatrix} 0 & -\partial_3 & \partial_2 \\ \partial_3 & 0 & -\partial_1 \\ -\partial_2 & \partial_1 & 0 \end{bmatrix} \cdot \begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix} = \begin{bmatrix} 0 \cdot F_1 - \partial_3 F_2 + \partial_2 F_3 \\ \partial_3 F_1 + 0 \cdot F_2 - \partial_1 F_3 \\ -\partial_2 F_1 + \partial_1 F_2 + 0 \cdot F_3 \end{bmatrix} = \begin{bmatrix} \partial_2 F_3 - \partial_3 F_2 \\ \partial_3 F_1 - \partial_1 F_3 \\ \partial_1 F_2 - \partial_2 F_1 \end{bmatrix}, \\ \nabla \times (\nabla u) &= \begin{bmatrix} \partial_2 \partial_3 u - \partial_3 \partial_2 u \\ \partial_3 \partial_1 u - \partial_1 \partial_3 u \\ \partial_1 \partial_2 u - \partial_2 \partial_1 u \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \\ \nabla \cdot (\nabla \times \mathbf{F}) &= \partial_1 (\partial_2 F_3 - \partial_3 F_2) + \partial_2 (\partial_3 F_1 - \partial_1 F_3) + \partial_3 (\partial_1 F_2 - \partial_2 F_1), \\ \nabla \cdot (\nabla \times \mathbf{F}) &= \color{red}{\partial_1 \partial_2 F_3} - \color{blue}{\partial_1 \partial_3 F_2} + \color{green}{\partial_2 \partial_3 F_1} - \color{red}{\partial_2 \partial_1 F_3} + \color{blue}{\partial_3 \partial_1 F_2} - \color{green}{\partial_3 \partial_2 F_1} = 0. \end{aligned}$$

□

Fact 3.2.9 (Stokes' theorem). *Let τ be an oriented smooth $(p+1)$ -simplex and $\omega \in \Omega^p(\overline{\Omega})$, with the boundary orientation of Definition 3.1.3.*

$$\int_{\tau} d\omega = \int_{\partial_{p+1}\tau} \omega.$$

Proof. [14, Section 4.1]. □

Example 3.2.10 (Stokes' theorem on the reference triangle). Let $\omega = xdy$ and orient $\hat{T} = [\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2]$ counterclockwise.

$$\begin{aligned} d\omega &= d(xdy) = dx \wedge dy, & \int_{\hat{T}} d\omega &= \int_0^1 \int_0^{1-y} 1 dx dy = \int_0^1 (1-y) dy = \left[y - \frac{y^2}{2} \right]_0^1 = \frac{1}{2}, \\ \mathbf{r}_{01}(t) &= \begin{bmatrix} t \\ 0 \end{bmatrix}, & \mathbf{r}_{12}(t) &= \begin{bmatrix} 1-t \\ t \end{bmatrix}, & \mathbf{r}_{20}(t) &= \begin{bmatrix} 0 \\ 1-t \end{bmatrix} \quad (0 \leq t \leq 1), \\ \int_{e_{01}} \omega &= \int_0^1 t \cdot 0 dt = 0, & \int_{e_{12}} \omega &= \int_0^1 (1-t) \cdot 1 dt = \left[t - \frac{t^2}{2} \right]_0^1 = \frac{1}{2}, & \int_{e_{20}} \omega &= \int_0^1 0 \cdot (-1) dt = 0, \\ \hat{T} = [\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2] &\xrightarrow{3.1.3} \partial_2 \hat{T} = e_{12} + e_{20} + e_{01}, & \int_{\partial_2 \hat{T}} \omega &= \frac{1}{2} + 0 + 0 = \frac{1}{2} \xrightarrow{3.2.9} \int_{\hat{T}} d\omega = \frac{1}{2}. \end{aligned}$$

Definition 3.2.11 (de Rham integration map). For $\omega \in \Omega^p(\overline{\Omega})$ and $\sigma \in K_p$, define [13, Section 5] [14, Sections 5.2 and 5.6]

$$R^p : \Omega^p(\overline{\Omega}) \longrightarrow C^p(K), \quad (R^p \omega)(\sigma) := \int_{\sigma} \omega,$$

$$(R^0 u)(v) = u(v), \quad (R^1 \mathbf{F})(e) = \int_e \mathbf{F}^T \cdot \mathbf{t}_e ds, \quad (R^2 \mathbf{G})(f) = \int_f \mathbf{G}^T \cdot \mathbf{n}_f dS, \quad (R^3 q)(T) = \int_T q d\mathbf{x},$$

where $\mathbf{t}_e$ and $\mathbf{n}_f$ are the oriented unit tangent and unit normal.

Proposition 3.2.12 (The integration map is a cochain map). For $0 \leq p < d$,

$$\delta^p \circ R^p = R^{p+1} \circ d.$$

Proof. Let $\omega \in \Omega^p(\overline{\Omega})$ and $\tau \in K_{p+1}$; the coboundary evaluates $R^p \omega$ on $\partial_{p+1} \tau$, and Stokes' theorem converts that boundary integral into the integral of $d\omega$ over τ .

$$\begin{aligned} \omega \in \Omega^p(\overline{\Omega}), \quad \tau \in K_{p+1} &\xrightarrow{3.2.2} (\delta^p R^p \omega)(\tau) = (R^p \omega)(\partial_{p+1} \tau) \xrightarrow{3.2.11} \int_{\partial_{p+1} \tau} \omega, \\ &\int_{\partial_{p+1} \tau} \omega \xrightarrow{3.2.9} \int_{\tau} d\omega \xrightarrow{3.2.11} (R^{p+1} d\omega)(\tau), \\ \forall \tau \in K_{p+1} : (\delta^p R^p \omega)(\tau) &= (R^{p+1} d\omega)(\tau) \implies \delta^p R^p \omega = R^{p+1} d\omega. \end{aligned}$$

□

Fact 3.2.13 (de Rham theorem for a triangulation). *Let Ω be a bounded polyhedral domain and let K be a finite triangulation of $\overline{\Omega}$.*

$$\begin{aligned} d\omega = 0 &\xrightarrow{3.2.12} \delta^p (R^p \omega) = R^{p+1} (d\omega) = 0, & \omega = d\eta &\xrightarrow{3.2.12} R^p \omega = R^p (d\eta) = \delta^{p-1} (R^{p-1} \eta), \\ [R^p] : H_{\text{dR}}^p(\Omega) &\longrightarrow H^p(K), & [\omega] &\longmapsto [R^p \omega], & [R^p] : H_{\text{dR}}^p(\Omega) &\xrightarrow{\sim} H^p(K). \end{aligned}$$

Proof. [14, Section 5.6]. □

4. WHITNEY FORMS

Whitney reconstruction identifies simplicial cochains with the lowest-order conforming finite-element de Rham complex.

4.1. Local forms.

Definition 4.1.1 (Barycentric coordinate matrix). Let $T = [\mathbf{v}_0, \dots, \mathbf{v}_d] \subset \mathbb{R}^d$ be nondegenerate and positively oriented [10, Sections 2–3].

$$\mathbb{B}_T = \begin{bmatrix} \mathbf{v}_0 & \mathbf{v}_1 & \cdots & \mathbf{v}_d \\ 1 & 1 & \cdots & 1 \end{bmatrix}, \quad \mathbf{L}_T(\mathbf{x}) = \begin{bmatrix} \lambda_0(\mathbf{x}) \\ \lambda_1(\mathbf{x}) \\ \vdots \\ \lambda_d(\mathbf{x}) \end{bmatrix}, \quad \mathbb{B}_T \cdot \mathbf{L}_T(\mathbf{x}) = \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix}, \quad \mathbf{L}_T(\mathbf{x}) = \mathbb{B}_T^{-1} \cdot \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix},$$

$$\mathbf{x} = \sum_{i=0}^d \lambda_i(\mathbf{x}) \cdot \mathbf{v}_i, \quad \sum_{i=0}^d \lambda_i(\mathbf{x}) = 1, \quad \lambda_i(\mathbf{v}_j) = \delta_{ij}, \quad \sum_{i=0}^d \nabla \lambda_i = \mathbf{0},$$

$$\mathbb{B}_T^{-1} = \begin{bmatrix} (\nabla \lambda_0)^\top & \lambda_0(\mathbf{0}) \\ (\nabla \lambda_1)^\top & \lambda_1(\mathbf{0}) \\ \vdots & \vdots \\ (\nabla \lambda_d)^\top & \lambda_d(\mathbf{0}) \end{bmatrix}.$$

Example 4.1.2 (Reference-simplex coordinate matrices). Use the reference triangle of Example 2.2.5 and the reference tetrahedron with vertices $\mathbf{0}, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$.

$$\xrightarrow{4.1.1} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} \lambda_0 \\ \lambda_1 \\ \lambda_2 \end{bmatrix} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}, \quad \begin{bmatrix} \lambda_0 \\ \lambda_1 \\ \lambda_2 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 1-x-y \\ x \\ y \end{bmatrix},$$

$$G_{\hat{T}}^{(2)} = [\nabla \lambda_0 \quad \nabla \lambda_1 \quad \nabla \lambda_2] = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix},$$

$$\xrightarrow{4.1.1} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} \lambda_0 \\ \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}, \quad \begin{bmatrix} \lambda_0 \\ \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} = \begin{bmatrix} -1 & -1 & -1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} 1-x-y-z \\ x \\ y \\ z \end{bmatrix},$$

$$G_{\hat{T}}^{(3)} = [\nabla \lambda_0 \quad \nabla \lambda_1 \quad \nabla \lambda_2 \quad \nabla \lambda_3] = \begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}.$$

Definition 4.1.3 (Whitney form). Let $\sigma = [\mathbf{v}_{i_0}, \dots, \mathbf{v}_{i_p}]$ be an oriented p -face [10, Sections 4–5].

$$\phi_\sigma|_T = p! \cdot \sum_{m=0}^p (-1)^m \lambda_{i_m, T} d\lambda_{i_0, T} \wedge \cdots \wedge \widehat{d\lambda_{i_m, T}} \wedge \cdots \wedge d\lambda_{i_p, T} \quad (\sigma \subset T), \quad \phi_\sigma|_T = 0 \quad (\sigma \not\subset T),$$

$$f = T^+ \cap T^-, \quad \mathbf{v}_i \in f \implies \lambda_{i, T^+}|_f = \lambda_{i, T^-}|_f, \quad d_f(\lambda_{i, T^+}|_f) = d_f(\lambda_{i, T^-}|_f),$$

$$f = T^+ \cap T^-, \quad \sigma \not\subset f \implies \text{tr}_f(\phi_\sigma|_{T^+}) = \text{tr}_f(\phi_\sigma|_{T^-}) = 0,$$

$$f = T^+ \cap T^- \implies \text{tr}_f(\phi_\sigma|_{T^+}) = \text{tr}_f(\phi_\sigma|_{T^-}), \quad \bar{\sigma} = -\sigma \implies \phi_{\bar{\sigma}} = -\phi_\sigma.$$

Theorem 4.1.4 (Whitney degree-of-freedom duality). Choose one orientation for every geometric p -simplex and denote the resulting basis by K_p [10, 13, Sections 4–5].

$$\int_\tau \phi_\sigma = \delta_{\tau\sigma} = \begin{cases} 1, & \tau = \sigma, \\ 0, & \tau \neq \sigma, \end{cases} \quad \sigma, \tau \in K_p, \quad \dim \text{span}\{\phi_\sigma : \sigma \in K_p\} = |K_p|.$$

Proof. Fix $\sigma = [\mathbf{v}_{i_0}, \dots, \mathbf{v}_{i_p}]$, $\tau \in K_p$, and $T \in K_d$ with $\tau \subset T$; every barycentric factor of ϕ_σ vanishes on a face missing its vertex, and on σ itself the trace is the volume form.

$$\begin{aligned}
\sigma \not\subset T &\xrightarrow{4.1.3} \phi_\sigma|_T = 0 \implies \text{tr}_\tau(\phi_\sigma|_T) = 0, \\
\sigma \subset T, \quad \tau \neq \sigma &\implies \exists r \in \{0, \dots, p\} : \mathbf{v}_{i_r} \notin \tau \implies \lambda_{i_r, T}|_\tau = 0, \quad d_\tau(\lambda_{i_r, T}|_\tau) = 0 \xrightarrow{4.1.3} \text{tr}_\tau(\phi_\sigma|_T) = 0, \\
\tau \neq \sigma, \quad T \in K_d, \quad \tau \subset T &\implies \text{tr}_\tau(\phi_\sigma|_T) = 0 \implies \int_\tau \phi_\sigma = 0, \\
\tau = \sigma \subset T, \quad t_j = \lambda_{i_j, T}|_\sigma \quad (1 \leq j \leq p) &\implies \lambda_{i_0, T}|_\sigma = 1 - \sum_{j=1}^p t_j, \quad d_\sigma(\lambda_{i_0, T}|_\sigma) = - \sum_{j=1}^p dt_j, \\
&\xrightarrow{4.1.3} \text{tr}_\sigma(\phi_\sigma|_T) = p! dt_1 \wedge \dots \wedge dt_p \implies \int_\sigma \phi_\sigma = p! \int_{\{t_j \geq 0, \sum_{j=1}^p t_j \leq 1\}} dt_1 \dots dt_p = p! \cdot \frac{1}{p!} = 1, \\
\sum_{\sigma \in K_p} c_\sigma \phi_\sigma = 0 &\implies 0 = \int_\tau \sum_{\sigma \in K_p} c_\sigma \phi_\sigma = \sum_{\sigma \in K_p} c_\sigma \delta_{\tau\sigma} = c_\tau \implies (c_\sigma)_{\sigma \in K_p} = \mathbf{0}.
\end{aligned}$$

□

Example 4.1.5 (Complete Whitney edge matrix on the reference triangle). Order $e_{01} = [\mathbf{v}_0, \mathbf{v}_1]$, $e_{12} = [\mathbf{v}_1, \mathbf{v}_2]$, and $e_{20} = [\mathbf{v}_2, \mathbf{v}_0]$.

$$\begin{aligned}
\mathbf{w}_{01} &= \lambda_0 \cdot \nabla \lambda_1 - \lambda_1 \cdot \nabla \lambda_0 = \begin{bmatrix} (1-x-y) \cdot 1 - x \cdot (-1) \\ (1-x-y) \cdot 0 - x \cdot (-1) \end{bmatrix} = \begin{bmatrix} 1-y \\ x \end{bmatrix}, \\
\mathbf{w}_{12} &= \lambda_1 \cdot \nabla \lambda_2 - \lambda_2 \cdot \nabla \lambda_1 = \begin{bmatrix} x \cdot 0 - y \cdot 1 \\ x \cdot 1 - y \cdot 0 \end{bmatrix} = \begin{bmatrix} -y \\ x \end{bmatrix}, \quad \mathbf{w}_{20} = \lambda_2 \cdot \nabla \lambda_0 - \lambda_0 \cdot \nabla \lambda_2 = \begin{bmatrix} -y \\ x-1 \end{bmatrix}, \\
\mathbf{r}_{01}(t) &= \begin{bmatrix} t \\ 0 \end{bmatrix}, \quad \mathbf{r}_{12}(t) = \begin{bmatrix} 1-t \\ t \end{bmatrix}, \quad \mathbf{r}_{20}(t) = \begin{bmatrix} 0 \\ 1-t \end{bmatrix}, \quad \mathbf{r}'_{01}(t) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{r}'_{12}(t) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \quad \mathbf{r}'_{20}(t) = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \\
\left[\int_{e_a} \mathbf{w}_b^\top \cdot d\mathbf{r} \right]_{a,b \in \{01,12,20\}} &= \begin{bmatrix} \int_0^1 1 dt & \int_0^1 0 dt & \int_0^1 0 dt \\ \int_0^1 0 dt & \int_0^1 (t+1-t) dt & \int_0^1 0 dt \\ \int_0^1 0 dt & \int_0^1 0 dt & \int_0^1 1 dt \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{4.1.4} \int_{e_a} \mathbf{w}_b^\top \cdot d\mathbf{r} = \delta_{ab}, \\
\nabla \times \mathbf{w}_{01} &= \partial_x(x) - \partial_y(1-y) = 2, \quad \nabla \times \mathbf{w}_{12} = \partial_x(x) - \partial_y(-y) = 2, \quad \nabla \times \mathbf{w}_{20} = \partial_x(x-1) - \partial_y(-y) = 2, \\
|\hat{T}| &= \frac{1}{2} \implies \int_{\hat{T}} \nabla \times \mathbf{w}_{ij} dx dy = 2 \cdot \frac{1}{2} = 1, \quad ij \in \{01, 12, 20\}.
\end{aligned}$$

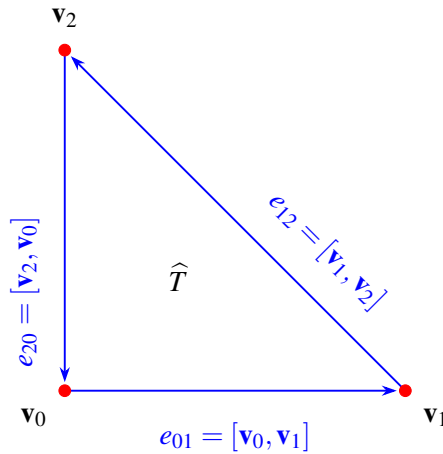


FIGURE 5. Oriented reference-triangle edges and Whitney circulation degrees of freedom.

Example 4.1.6 (Triangle incidence matrices). Let $\mathbf{u} = [u_0, u_1, u_2]^\top$ and use the orientations of Figure 5.

$$\begin{aligned} \begin{bmatrix} u_1 - u_0 \\ u_2 - u_1 \\ u_0 - u_2 \end{bmatrix} &= \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & -1 \end{bmatrix} \cdot \begin{bmatrix} u_0 \\ u_1 \\ u_2 \end{bmatrix} = D_{\hat{T}}^0 \cdot \mathbf{u}, \quad D_{\hat{T}}^1 = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}, \\ D_{\hat{T}}^1 \cdot D_{\hat{T}}^0 &= \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & -1 \end{bmatrix} = \begin{bmatrix} -1+0+1 & 1-1+0 & 0+1-1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}, \\ D_{\hat{T}}^0 &= [\partial_1]^\top, \quad D_{\hat{T}}^1 = [\partial_2]^\top \xrightarrow{3.2.3} D_{\hat{T}}^1 \cdot D_{\hat{T}}^0 = ([\partial_1] \cdot [\partial_2])^\top \xrightarrow{3.1.7} \mathbf{0}. \end{aligned}$$

4.2. Assembled spaces.

Definition 4.2.1 (Lowest-order local and assembled spaces). Let K be a tetrahedral triangulation of a polyhedral domain $\Omega \subset \mathbb{R}^3$ [12, 11, 13].

$$\begin{cases} \mathcal{P}_1(T) = \{a_0 + \mathbf{a}^\top \cdot \mathbf{x} : a_0 \in \mathbb{R}, \mathbf{a} \in \mathbb{R}^3\}, \\ \mathcal{N}_0(T) = \{\mathbf{a} + \mathbf{b} \times \mathbf{x} : \mathbf{a}, \mathbf{b} \in \mathbb{R}^3\}, \\ \mathcal{RT}_0(T) = \{\mathbf{a} + c \cdot \mathbf{x} : \mathbf{a} \in \mathbb{R}^3, c \in \mathbb{R}\}, \\ \mathcal{P}_0(T) = \{c : c \in \mathbb{R}\}, \end{cases} \quad \begin{cases} V_h^0 = \{u_h \in H^1(\Omega) : u_h|_T \in \mathcal{P}_1(T) \ (T \in K_3)\}, \\ V_h^1 = \{\mathbf{v}_h \in H(\nabla \times, \Omega) : \mathbf{v}_h|_T \in \mathcal{N}_0(T) \ (T \in K_3)\}, \\ V_h^2 = \{\mathbf{q}_h \in H(\nabla \cdot, \Omega) : \mathbf{q}_h|_T \in \mathcal{RT}_0(T) \ (T \in K_3)\}, \\ V_h^3 = \{w_h \in L^2(\Omega) : w_h|_T \in \mathcal{P}_0(T) \ (T \in K_3)\}, \end{cases}$$

$$\begin{aligned} H(\nabla \times, \Omega) &= \{\mathbf{v} \in L^2(\Omega; \mathbb{R}^3) : \nabla \times \mathbf{v} \in L^2(\Omega; \mathbb{R}^3)\}, \quad H(\nabla \cdot, \Omega) = \{\mathbf{q} \in L^2(\Omega; \mathbb{R}^3) : \nabla \cdot \mathbf{q} \in L^2(\Omega)\}, \\ \mathbf{n}_{T^+} \times \mathbf{v}_h|_{T^+} &= -\mathbf{n}_{T^-} \times \mathbf{v}_h|_{T^-}, \quad \mathbf{n}_{T^+}^\top \cdot \mathbf{q}_h|_{T^+} = -\mathbf{n}_{T^-}^\top \cdot \mathbf{q}_h|_{T^-} \end{aligned}$$

on every interior face $T^+ \cap T^-$, with outward normals $\mathbf{n}_{T^+} = -\mathbf{n}_{T^-}$.

Definition 4.2.2 (Oriented geometric degrees of freedom). Let $\mathbf{v}$, e , f , and T be an oriented vertex, edge, face, and tetrahedron.

$$\begin{cases} R_{\mathbf{v}}^0(u_h) = u_h(\mathbf{v}), \\ R_e^1(\mathbf{v}_h) = \int_e \mathbf{v}_h^\top \cdot \mathbf{t}_e ds, \\ R_f^2(\mathbf{q}_h) = \int_f \mathbf{q}_h^\top \cdot \mathbf{n}_f dS, \\ R_T^3(w_h) = \int_T w_h d\mathbf{x}, \end{cases} \quad \begin{bmatrix} \dim \mathcal{P}_1(T) \\ \dim \mathcal{N}_0(T) \\ \dim \mathcal{RT}_0(T) \\ \dim \mathcal{P}_0(T) \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \\ 4 \\ 1 \end{bmatrix}.$$

Proposition 4.2.3 (Local exactness and unisolvence). *For every tetrahedron T , the geometric degrees of freedom of Definition 4.2.2 are unisolvent and [12, 11, 13]*

$$0 \longrightarrow \mathbb{R} \xrightarrow{c \mapsto c} \mathcal{P}_1(T) \xrightarrow{\nabla} \mathcal{N}_0(T) \xrightarrow{\nabla \times} \mathcal{RT}_0(T) \xrightarrow{\nabla \cdot} \mathcal{P}_0(T) \longrightarrow 0$$

is exact.

Proof. Write $u = a_0 + \mathbf{a}^\top \cdot \mathbf{x}$, $\mathbf{v} = \mathbf{a} + \mathbf{b} \times \mathbf{x}$, and $\mathbf{q} = \mathbf{a} + c \cdot \mathbf{x}$; ranges and kernels line up degree by degree, and vanishing degrees of freedom force each field to zero.

$$\begin{aligned} \xrightarrow{3.2.8} \nabla u &= \mathbf{a}, \quad \nabla \times \mathbf{v} = \begin{bmatrix} \partial_y(b_1y - b_2x) - \partial_z(b_3x - b_1z) \\ \partial_z(b_2z - b_3y) - \partial_x(b_1y - b_2x) \\ \partial_x(b_3x - b_1z) - \partial_y(b_2z - b_3y) \end{bmatrix} = 2\mathbf{b}, \\ \nabla \cdot \mathbf{q} &= \partial_x(a_1 + cx) + \partial_y(a_2 + cy) + \partial_z(a_3 + cz) = 3c, \\ \ker(\nabla) &= \mathbb{R}, \quad \text{ran}(\nabla) = \{\mathbf{a} : \mathbf{a} \in \mathbb{R}^3\} = \ker(\nabla \times), \\ \text{ran}(\nabla \times) &= \{2\mathbf{b} : \mathbf{b} \in \mathbb{R}^3\} = \mathbb{R}^3 = \ker(\nabla \cdot), \quad \text{ran}(\nabla \cdot) = \mathcal{P}_0(T), \\ (R_e^1(\mathbf{v}) = 0 \ (e \subset \partial T)) &\implies 0 = \sum_{e \subset \partial f} \int_e \mathbf{v}^\top \cdot \mathbf{t}_e ds \xrightarrow{3.2.9} 0 = \int_f (\nabla \times \mathbf{v})^\top \cdot \mathbf{n}_f dS = 2|f| \cdot \mathbf{b}^\top \cdot \mathbf{n}_f \quad (f \subset \partial T), \end{aligned}$$

$$\begin{aligned}
& \text{span}\{\mathbf{n}_{f_1}, \mathbf{n}_{f_2}, \mathbf{n}_{f_3}\} = \mathbb{R}^3 \implies \mathbf{b} = \mathbf{0} \implies \mathbf{v} = \mathbf{a}, \\
& 0 = R_{[\mathbf{v}_0, \mathbf{v}_j]}^1(\mathbf{a}) = \mathbf{a}^\top \cdot (\mathbf{v}_j - \mathbf{v}_0) \quad (1 \leq j \leq 3) \implies \mathbf{a} = \mathbf{0}, \\
& (R_f^2(\mathbf{q}) = 0 \quad (f \subset \partial T)) \implies 0 = \sum_{f \subset \partial T} \int_f \mathbf{q}^\top \cdot \mathbf{n}_f dS \xrightarrow{1.2.1} 0 = \int_T \nabla \cdot \mathbf{q} d\mathbf{x} = 3c|T| \implies c = 0, \\
& c = 0, \quad 0 = R_f^2(\mathbf{a}) = |f| \cdot \mathbf{a}^\top \cdot \mathbf{n}_f \quad (f \subset \partial T) \implies \mathbf{a} = \mathbf{0}, \\
& (u(\mathbf{v}_i) = 0 \quad (0 \leq i \leq 3)) \xrightarrow{4.1.1} u = \sum_{i=0}^3 u(\mathbf{v}_i) \lambda_{i,T} = 0, \quad w \in \mathcal{P}_0(T), \quad R_T^3(w) = w|T| = 0 \implies w = 0, \\
& \mathcal{R}_T^0: \mathcal{P}_1(T) \longrightarrow \mathbb{R}^4, \quad \mathcal{R}_T^1: \mathcal{N}_0(T) \longrightarrow \mathbb{R}^6, \quad \mathcal{R}_T^2: \mathcal{RT}_0(T) \longrightarrow \mathbb{R}^4, \quad \mathcal{R}_T^3: \mathcal{P}_0(T) \longrightarrow \mathbb{R}, \\
& \begin{bmatrix} \ker(\mathcal{R}_T^0) \\ \ker(\mathcal{R}_T^1) \\ \ker(\mathcal{R}_T^2) \\ \ker(\mathcal{R}_T^3) \end{bmatrix} = \begin{bmatrix} \{0\} \\ \{0\} \\ \{0\} \\ \{0\} \end{bmatrix}, \quad \begin{bmatrix} \dim \mathcal{P}_1(T) \\ \dim \mathcal{N}_0(T) \\ \dim \mathcal{RT}_0(T) \\ \dim \mathcal{P}_0(T) \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \\ 4 \\ 1 \end{bmatrix} \xrightarrow{4.2.2} \begin{cases} \mathcal{R}_T^0: \mathcal{P}_1(T) \xrightarrow{\cong} \mathbb{R}^4, \\ \mathcal{R}_T^1: \mathcal{N}_0(T) \xrightarrow{\cong} \mathbb{R}^6, \\ \mathcal{R}_T^2: \mathcal{RT}_0(T) \xrightarrow{\cong} \mathbb{R}^4, \\ \mathcal{R}_T^3: \mathcal{P}_0(T) \xrightarrow{\cong} \mathbb{R}. \end{cases}
\end{aligned}$$

□

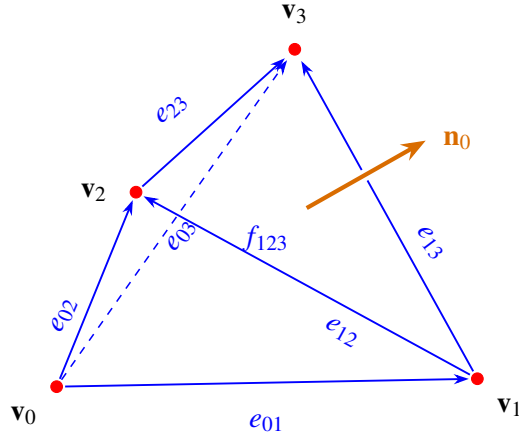


FIGURE 6. Oriented reference-tetrahedron edges and one outward face normal.

Example 4.2.4 (Complete Nedgelec and Raviart–Thomas matrices). On the tetrahedron of Example 4.1.2, order the edges by 01, 02, 03, 12, 13, 23 and the outward faces by f_0, f_1, f_2, f_3 .

$$\mathbf{w}_{01} = \begin{bmatrix} 1-y-z \\ x \\ x \end{bmatrix}, \quad \mathbf{w}_{02} = \begin{bmatrix} y \\ 1-x-z \\ y \end{bmatrix}, \quad \mathbf{w}_{03} = \begin{bmatrix} z \\ z \\ 1-x-y \end{bmatrix}, \quad \mathbf{w}_{12} = \begin{bmatrix} -y \\ x \\ 0 \end{bmatrix}, \quad \mathbf{w}_{13} = \begin{bmatrix} -z \\ 0 \\ x \end{bmatrix}, \quad \mathbf{w}_{23} = \begin{bmatrix} 0 \\ -z \\ y \end{bmatrix},$$

$$\mathbf{r}_{ij}(t) = (1-t) \cdot \mathbf{v}_i + t \cdot \mathbf{v}_j, \quad d\mathbf{r} = (\mathbf{v}_j - \mathbf{v}_i) dt, \quad 0 \leq t \leq 1,$$

$$ab \neq ij \implies \int_{e_{ab}} \mathbf{w}_{ij}^\top \cdot d\mathbf{r} = \int_0^1 0 dt = 0, \quad \int_{e_{0j}} \mathbf{w}_{0j}^\top \cdot d\mathbf{r} = \int_0^1 1 dt = 1 \quad (1 \leq j \leq 3),$$

$$\int_{e_{ij}} \mathbf{w}_{ij}^\top \cdot d\mathbf{r} = \int_0^1 (t+1-t) dt = 1 \quad (1 \leq i < j \leq 3) \implies \left[\int_{e_{ab}} \mathbf{w}_{ij}^\top \cdot d\mathbf{r} \right]_{ab,ij} = I_6 \xleftrightarrow{4.1.4} \delta_{ab,ij},$$

$$C_{\hat{T}} := \begin{bmatrix} \nabla \times \mathbf{w}_{01} & \nabla \times \mathbf{w}_{02} & \nabla \times \mathbf{w}_{03} & \nabla \times \mathbf{w}_{12} & \nabla \times \mathbf{w}_{13} & \nabla \times \mathbf{w}_{23} \end{bmatrix} = \begin{bmatrix} 0 & 2 & -2 & 0 & 0 & 2 \\ -2 & 0 & 2 & 0 & -2 & 0 \\ 2 & -2 & 0 & 2 & 0 & 0 \end{bmatrix},$$

$$\begin{aligned}
|\widehat{T}| &= \frac{1}{6}, \quad \mathbf{q}_i(\mathbf{x}) = \frac{\mathbf{x} - \mathbf{v}_i}{3|\widehat{T}|} = 2 \cdot (\mathbf{x} - \mathbf{v}_i), \quad 0 \leq i \leq 3, \\
\mathbf{n}_0 &= \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad |f_0| = \frac{\sqrt{3}}{2}, \quad \mathbf{n}_1 = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{n}_2 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}, \quad \mathbf{n}_3 = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}, \quad |f_1| = |f_2| = |f_3| = \frac{1}{2}, \\
A_F &:= \begin{bmatrix} |f_0| \mathbf{n}_0^\top \\ |f_1| \mathbf{n}_1^\top \\ |f_2| \mathbf{n}_2^\top \\ |f_3| \mathbf{n}_3^\top \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & 0 & 0 \\ 0 & -\frac{1}{2} & 0 \\ 0 & 0 & -\frac{1}{2} \end{bmatrix}, \\
A_F \cdot C_{\widehat{T}} &= \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & 0 & 0 \\ 0 & -\frac{1}{2} & 0 \\ 0 & 0 & -\frac{1}{2} \end{bmatrix} \cdot \begin{bmatrix} 0 & 2 & -2 & 0 & 0 & 2 \\ -2 & 0 & 2 & 0 & -2 & 0 \\ 2 & -2 & 0 & 2 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 & -1 & 1 \\ 0 & -1 & 1 & 0 & 0 & -1 \\ 1 & 0 & -1 & 0 & 1 & 0 \\ -1 & 1 & 0 & -1 & 0 & 0 \end{bmatrix} = D_{\widehat{T}}^1, \\
i \neq j, \quad \mathbf{x}, \mathbf{v}_i \in f_j &\implies (\mathbf{x} - \mathbf{v}_i)^\top \cdot \mathbf{n}_j = 0 \implies \int_{f_j} \mathbf{q}_i^\top \cdot \mathbf{n}_j dS = 0, \\
\left[\int_{f_j} \mathbf{q}_i^\top \cdot \mathbf{n}_j dS \right]_{j,i=0}^3 &= \begin{bmatrix} \frac{2}{\sqrt{3}} \cdot \frac{\sqrt{3}}{2} & 0 & 0 & 0 \\ 0 & 2 \cdot \frac{1}{2} & 0 & 0 \\ 0 & 0 & 2 \cdot \frac{1}{2} & 0 \\ 0 & 0 & 0 & 2 \cdot \frac{1}{2} \end{bmatrix} = I_4, \\
\nabla \cdot \mathbf{q}_i &= 2\partial_x(x) + 2\partial_y(y) + 2\partial_z(z) = 6 \xrightarrow{1.2.1} \int_{\widehat{T}} \nabla \cdot \mathbf{q}_i d\mathbf{x} = 6 \cdot \frac{1}{6} = 1 = \sum_{j=0}^3 \int_{f_j} \mathbf{q}_i^\top \cdot \mathbf{n}_j dS.
\end{aligned}$$

Example 4.2.5 (Tetrahedral incidence matrices). Order vertices by 0, 1, 2, 3, edges by 01, 02, 03, 12, 13, 23, and outward faces by f_0, f_1, f_2, f_3 .

$$D_{\widehat{T}}^0 = \begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix}, \quad D_{\widehat{T}}^1 = \begin{bmatrix} 0 & 0 & 0 & 1 & -1 & 1 \\ 0 & -1 & 1 & 0 & 0 & -1 \\ 1 & 0 & -1 & 0 & 1 & 0 \\ -1 & 1 & 0 & -1 & 0 & 0 \end{bmatrix}, \quad D_{\widehat{T}}^2 = [1 \quad 1 \quad 1 \quad 1],$$

$$D_{\widehat{T}}^1 \cdot D_{\widehat{T}}^0 = \begin{bmatrix} 0 & 0 & 0 & 1 & -1 & 1 \\ 0 & -1 & 1 & 0 & 0 & -1 \\ 1 & 0 & -1 & 0 & 1 & 0 \\ -1 & 1 & 0 & -1 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

$$D_{\widehat{T}}^2 \cdot D_{\widehat{T}}^1 = [1 \quad 1 \quad 1 \quad 1] \cdot \begin{bmatrix} 0 & 0 & 0 & 1 & -1 & 1 \\ 0 & -1 & 1 & 0 & 0 & -1 \\ 1 & 0 & -1 & 0 & 1 & 0 \\ -1 & 1 & 0 & -1 & 0 & 0 \end{bmatrix} = [0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0],$$

$$\mathbb{R}^4 \xrightarrow{D_{\widehat{T}}^0} \mathbb{R}^6 \xrightarrow{D_{\widehat{T}}^1} \mathbb{R}^4 \xrightarrow{D_{\widehat{T}}^2} \mathbb{R}, \quad D_{\widehat{T}}^{p+1} \cdot D_{\widehat{T}}^p = 0 \quad (0 \leq p \leq 1) \xrightarrow{3.1.7}$$

$$0 \longrightarrow \mathbb{R} \xrightarrow{c \mapsto c \cdot [1 \quad 1 \quad 1 \quad 1]^\top} \mathbb{R}^4 \xrightarrow{D_{\widehat{T}}^0} \mathbb{R}^6 \xrightarrow{D_{\widehat{T}}^1} \mathbb{R}^4 \xrightarrow{D_{\widehat{T}}^2} \mathbb{R} \longrightarrow 0.$$

4.3. Commuting interpolation.

Definition 4.3.1 (Reconstruction, integration, and interpolation). Let K_p carry fixed orientations, let $c \in C^p(K)$, and assume every displayed trace exists [10, 13].

$$W_h^p(c) = \sum_{\sigma \in K_p} c(\sigma) \phi_\sigma, \quad W_h^p : C^p(K) \longrightarrow \mathcal{W}_h^p, \quad \mathcal{W}_h^p = W_h^p(C^p(K)),$$

$$(R_h^p \eta_h)(\sigma) = \int_\sigma \text{tr}_\sigma(\eta_h), \quad R_h^p : V_h^p \longrightarrow C^p(K), \quad I_h^p = W_h^p \cdot R_h^p, \quad I_h^p \omega = \sum_{\sigma \in K_p} \left(\int_\sigma \omega \right) \phi_\sigma.$$

Proposition 4.3.2 (Cochain–finite-element isomorphism). *For $0 \leq p \leq 3$, the three-dimensional vector proxy satisfies*

$$\mathcal{W}_h^p = V_h^p, \quad R_h^p \cdot W_h^p = \text{Id}_{C^p(K)}, \quad W_h^p \cdot R_h^p = \text{Id}_{V_h^p}.$$

Proof. Fix $T \in K_3$, $0 \leq p \leq 3$, and $K_p(T) := \{\sigma \in K_p : \sigma \subset T\}$; duality makes the local Whitney forms a basis, matching traces glue them into the assembled spaces, and unisolvence inverts the count.

$$\left[\int_\tau \phi_\sigma \right]_{\tau, \sigma \in K_p(T)} \xrightarrow{4.1.4} I_{|K_p(T)|}, \quad \dim \text{span}\{\phi_\sigma|_T : \sigma \in K_p(T)\} = |K_p(T)|.$$

With the vector proxy of Proposition 3.2.8, orient $e_{ij} = [\mathbf{v}_i, \mathbf{v}_j]$, orient f_i outward, and let f_i be opposite $\mathbf{v}_i$.

$$\phi_{\mathbf{v}_i}|_T \longleftrightarrow \lambda_{i,T} \in \mathcal{P}_1(T), \quad \phi_{e_{ij}}|_T \longleftrightarrow \mathbf{w}_{ij} = \lambda_{i,T} \cdot \nabla \lambda_{j,T} - \lambda_{j,T} \cdot \nabla \lambda_{i,T} \in \mathcal{N}_0(T),$$

$$\phi_{f_i}|_T \longleftrightarrow \mathbf{q}_i(\mathbf{x}) = \frac{\mathbf{x} - \mathbf{v}_i}{3|T|} \in \mathcal{RT}_0(T), \quad \phi_T|_T \longleftrightarrow \frac{1}{|T|} \in \mathcal{P}_0(T),$$

$$|K_0(T)| = 4, \quad |K_1(T)| = 6, \quad |K_2(T)| = 4, \quad |K_3(T)| = 1 \xrightarrow{4.2.3} \text{span}\{\phi_\sigma|_T\} = \begin{cases} \mathcal{P}_1(T), & p = 0, \\ \mathcal{N}_0(T), & p = 1, \\ \mathcal{RT}_0(T), & p = 2, \\ \mathcal{P}_0(T), & p = 3, \end{cases}$$

$$f = T^+ \cap T^-, \quad 0 \leq p \leq 2 \xrightarrow{4.1.3} \text{tr}_f(W_h^p(c)|_{T^+}) = \sum_{\substack{\sigma \in K_p \\ \sigma \subset f}} c(\sigma) \text{tr}_f(\phi_\sigma|_{T^+}) = \text{tr}_f(W_h^p(c)|_{T^-}),$$

$$\begin{cases} W_h^0(c)|_{T^+ \cap f} = W_h^0(c)|_{T^- \cap f}, \\ \mathbf{n}_{T^+} \times W_h^1(c)|_{T^+} = -\mathbf{n}_{T^-} \times W_h^1(c)|_{T^-}, \\ \mathbf{n}_{T^+}^\top \cdot W_h^2(c)|_{T^+} = -\mathbf{n}_{T^-}^\top \cdot W_h^2(c)|_{T^-} \end{cases} \xrightarrow{4.2.1} \mathcal{W}_h^p \subseteq V_h^p \quad (0 \leq p \leq 2),$$

$$p = 3, \quad W_h^3(c)|_T \in \mathcal{P}_0(T) \quad (T \in K_3) \xrightarrow{4.2.1} W_h^3(c) \in V_h^3 \implies \mathcal{W}_h^3 \subseteq V_h^3,$$

$$c \in C^p(K) \xrightarrow{4.1.4} (R_h^p \cdot W_h^p(c))(\tau) = \int_\tau \sum_{\sigma \in K_p} c(\sigma) \phi_\sigma = \sum_{\sigma \in K_p} c(\sigma) \delta_{\tau\sigma} = c(\tau) \implies R_h^p \cdot W_h^p = \text{Id}_{C^p(K)},$$

$$\eta_h \in V_h^p \implies R_h^p \cdot W_h^p \cdot R_h^p(\eta_h) = R_h^p(\eta_h) \xrightarrow{4.2.3} W_h^p \cdot R_h^p(\eta_h) = \eta_h \implies V_h^p \subseteq \mathcal{W}_h^p, \quad W_h^p \cdot R_h^p = \text{Id}_{V_h^p}. \quad \square$$

Theorem 4.3.3 (Incidence differentiation and commuting interpolation). *For $0 \leq p < d$, define $[\tau : \sigma] \in \{-1, 0, 1\}$ by $\partial_{p+1}\tau = \sum_{\sigma \in K_p} [\tau : \sigma] \sigma$ as in Definition 3.1.6; then [10, 13, 14]*

$$dW_h^p = W_h^{p+1} \delta^p, \quad d\phi_\sigma = \sum_{\tau \in K_{p+1}} [\tau : \sigma] \phi_\tau, \quad dI_h^p = I_h^{p+1} d,$$

$$\nabla I_h^0 u = I_h^1(\nabla u), \quad \nabla \times I_h^1 \mathbf{v} = I_h^2(\nabla \times \mathbf{v}), \quad \nabla \cdot I_h^2 \mathbf{q} = I_h^3(\nabla \cdot \mathbf{q}).$$

Proof. Fix $\sigma = [\mathbf{v}_{i_0}, \dots, \mathbf{v}_{i_p}] \subset T \in K_d$ and abbreviate $\lambda_i = \lambda_{i,T}$; on one cell the derivative of ϕ_σ is the sum of the Whitney forms of its cofaces, and matching traces glue the cellwise identity globally.

$$\begin{aligned}
\Omega_\sigma &= d\lambda_{i_0} \wedge \dots \wedge d\lambda_{i_p}, & \Omega_{\sigma,m} &= d\lambda_{i_0} \wedge \dots \wedge \widehat{d\lambda_{i_m}} \wedge \dots \wedge d\lambda_{i_p}, \\
d(\phi_\sigma|_T) &\xrightarrow{4.1.3} p! \cdot \sum_{m=0}^p (-1)^m d\lambda_{i_m} \wedge \Omega_{\sigma,m} = p! \cdot \sum_{m=0}^p \Omega_\sigma = (p+1)! \Omega_\sigma, \\
J_T &= \{j : \mathbf{v}_j \in T, \mathbf{v}_j \notin \sigma\}, & \tau_j &= [\mathbf{v}_j, \mathbf{v}_{i_0}, \dots, \mathbf{v}_{i_p}], \\
\sum_{j \in J_T} \phi_{\tau_j} &= (p+1)! \left[\left(\sum_{j \in J_T} \lambda_j \right) \Omega_\sigma + \sum_{m=0}^p (-1)^{m+1} \lambda_{i_m} \left(\sum_{j \in J_T} d\lambda_j \right) \wedge \Omega_{\sigma,m} \right], \\
\sum_{j \in J_T} \lambda_j &= 1 - \sum_{m=0}^p \lambda_{i_m}, & \sum_{j \in J_T} d\lambda_j &= - \sum_{m=0}^p d\lambda_{i_m}, & (-1)^m d\lambda_{i_m} \wedge \Omega_{\sigma,m} &= \Omega_\sigma, \\
\sum_{j \in J_T} \phi_{\tau_j} &= (p+1)! \left(1 - \sum_{m=0}^p \lambda_{i_m} + \sum_{m=0}^p \lambda_{i_m} \right) \Omega_\sigma = (p+1)! \Omega_\sigma = d(\phi_\sigma|_T), \\
\tau \mapsto -\tau &\implies [\tau : \sigma] \mapsto -[\tau : \sigma], & \phi_\tau \mapsto -\phi_\tau &\implies d\phi_\sigma|_T = \sum_{\substack{\tau \in K_{p+1} \\ \tau \subset T}} [\tau : \sigma] \phi_\tau|_T.
\end{aligned}$$

Matching traces kill the interface terms, so the cellwise derivative is the distributional one.

$$\begin{aligned}
\phi_\sigma &= W_h^p(\sigma^*) \in \mathcal{W}_h^p \xrightarrow{4.3.2} \phi_\sigma \in V_h^p, & f &= T^+ \cap T^- \xrightarrow{4.1.3} \text{tr}_f(\phi_\sigma|_{T^+}) = \text{tr}_f(\phi_\sigma|_{T^-}), \\
d_h \phi_\sigma|_T &:= d(\phi_\sigma|_T), & \eta &\in \Omega_c^{d-p-1}(\Omega), \\
\sum_{T \in K_d} \int_{\partial T} \text{tr}_{\partial T}(\phi_\sigma) \wedge \text{tr}_{\partial T}(\eta) &= \sum_{f=T^+ \cap T^-} \left[\int_f \text{tr}_f(\phi_\sigma|_{T^+}) \wedge \text{tr}_f(\eta) - \int_f \text{tr}_f(\phi_\sigma|_{T^-}) \wedge \text{tr}_f(\eta) \right] = 0, \\
\int_\Omega d_h \phi_\sigma \wedge \eta &= \sum_{T \in K_d} \int_T d(\phi_\sigma|_T) \wedge \eta \xrightarrow{3.2.9} (-1)^{p+1} \int_\Omega \phi_\sigma \wedge d\eta \implies d\phi_\sigma = d_h \phi_\sigma, \\
d\phi_\sigma|_T &= \sum_{\substack{\tau \in K_{p+1} \\ \tau \subset T}} [\tau : \sigma] \phi_\tau|_T \quad (T \in K_d) \implies d\phi_\sigma = \sum_{\tau \in K_{p+1}} [\tau : \sigma] \phi_\tau.
\end{aligned}$$

Interpolation commutes by composing with the de Rham map.

$$\begin{aligned}
dW_h^p(c) &= \sum_{\sigma \in K_p} c(\sigma) d\phi_\sigma = \sum_{\tau \in K_{p+1}} \left(\sum_{\sigma \in K_p} [\tau : \sigma] c(\sigma) \right) \phi_\tau \xrightarrow{3.2.2} W_h^{p+1}(\delta^p c), \\
dI_h^p \omega &\xrightarrow{4.3.1} dW_h^p R^p \omega = W_h^{p+1} \delta^p R^p \omega \xrightarrow{3.2.12} W_h^{p+1} R^{p+1} d\omega = I_h^{p+1} d\omega, \\
dI_h^p &= I_h^{p+1} d \xrightarrow{3.2.7} \begin{cases} \nabla I_h^0 u = I_h^1(\nabla u), \\ \nabla \times I_h^1 \mathbf{v} = I_h^2(\nabla \times \mathbf{v}), \\ \nabla \cdot I_h^2 \mathbf{q} = I_h^3(\nabla \cdot \mathbf{q}). \end{cases}
\end{aligned}$$

□

Example 4.3.4 (Complete commuting calculation on one triangle). On the reference triangle with the orientations of Figure 5, let $\mathbf{F}(x, y) = [2 - y, x + 3]^\top$.

$$\begin{aligned}
R_{e_{01}}^1(\mathbf{F}) &= \int_0^1 \begin{bmatrix} 2 \\ t+3 \end{bmatrix}^\top \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} dt = \int_0^1 2 dt = 2, & R_{e_{12}}^1(\mathbf{F}) &= \int_0^1 (-(2-t) + (4-t)) dt = 2, \\
R_{e_{20}}^1(\mathbf{F}) &= \int_0^1 \begin{bmatrix} 1+t \\ 3 \end{bmatrix}^\top \cdot \begin{bmatrix} 0 \\ -1 \end{bmatrix} dt = \int_0^1 (-3) dt = -3,
\end{aligned}$$

$$\begin{aligned}
I_h^1 \mathbf{F} &\xrightarrow{4.3.1} 2\mathbf{w}_{01} + 2\mathbf{w}_{12} - 3\mathbf{w}_{20} = 2 \begin{bmatrix} 1-y \\ x \end{bmatrix} + 2 \begin{bmatrix} -y \\ x \end{bmatrix} - 3 \begin{bmatrix} -y \\ x-1 \end{bmatrix} = \begin{bmatrix} 2-y \\ x+3 \end{bmatrix} = \mathbf{F}, \\
\nabla \times \mathbf{F} &= \partial_x(x+3) - \partial_y(2-y) = 2, \quad \int_{\hat{T}} \nabla \times \mathbf{F} \, dx dy = 2 \cdot \frac{1}{2} = 1, \quad \int_{\partial_2 \hat{T}} \mathbf{F}^\top \cdot d\mathbf{r} = 2 + 2 - 3 = 1, \\
\nabla \times I_h^1 \mathbf{F} &\xrightarrow{4.3.3} I_h^2(\nabla \times \mathbf{F}) = 2.
\end{aligned}$$

Corollary 4.3.5 (Discrete cohomology). *Let $|K| = \overline{\Omega}$ be a finite triangulation [13, 14, 8].*

$$\begin{aligned}
V_h^0 &\xrightarrow{\nabla} V_h^1 \xrightarrow{\nabla \times} V_h^2 \xrightarrow{\nabla \cdot} V_h^3, \quad H^p(V_h^\bullet, d) \cong H^p(K) \cong H_{\text{dR}}^p(\Omega), \\
|K| \simeq \{*\} &\implies 0 \longrightarrow \mathbb{R} \longrightarrow V_h^0 \xrightarrow{\nabla} V_h^1 \xrightarrow{\nabla \times} V_h^2 \xrightarrow{\nabla \cdot} V_h^3 \longrightarrow 0 \quad \text{exact.}
\end{aligned}$$

Proof. Let $\iota_K : \mathbb{R} \rightarrow C^0(K)$ send c to the constant cochain; the Whitney maps are cochain isomorphisms, so discrete cohomology is simplicial cohomology, and contractibility empties it.

$$\begin{aligned}
R_h^p \cdot W_h^p &= \text{Id}, \quad W_h^p \cdot R_h^p = \text{Id} \xrightarrow{4.3.2} R_h^p = (W_h^p)^{-1} : V_h^p \longrightarrow C^p(K), \\
dW_h^p &= W_h^{p+1} \delta^p \xrightarrow{4.3.3} (V_h^\bullet, d) \cong (C^\bullet(K), \delta) \implies H^p(V_h^\bullet, d) \cong H^p(K), \\
|K| = \overline{\Omega} &\xrightarrow{3.2.13} H^p(K) \cong H_{\text{dR}}^p(\Omega) \implies H^p(V_h^\bullet, d) \cong H^p(K) \cong H_{\text{dR}}^p(\Omega), \\
|K| = \overline{\Omega} \simeq \{*\} &\implies H^0(K) \cong \mathbb{R}, \quad H^p(K) = 0 \quad (1 \leq p \leq 3), \\
(\iota_K c)(\mathbf{v}) &= c, \quad e = [\mathbf{v}_i, \mathbf{v}_j] \in K_1 \xrightarrow{3.2.2} (\delta^0 \iota_K c)(e) = c - c = 0 \implies \text{ran}(\iota_K) \subseteq \ker(\delta^0), \\
\mathcal{B}^0(K) = \{0\} &\xrightarrow{3.2.5} \dim \ker(\delta^0) = \dim H^0(K) = 1, \quad \ker(\iota_K) = \{0\} \implies \text{ran}(\iota_K) = \ker(\delta^0), \\
H^p(K) = 0 \quad (1 \leq p \leq 3) &\iff \begin{cases} \text{ran}(\delta^0) = \ker(\delta^1), \\ \text{ran}(\delta^1) = \ker(\delta^2), \\ \text{ran}(\delta^2) = C^3(K), \end{cases} \\
R_h^p = (W_h^p)^{-1}, \quad dW_h^p &= W_h^{p+1} \delta^p \xrightarrow{4.3.2, 4.3.3} W_h^p(\ker(\delta^p)) = \ker(d|_{V_h^p}), \quad W_h^{p+1}(\text{ran}(\delta^p)) = \text{ran}(d|_{V_h^p}), \\
\iota_h := W_h^0 \cdot \iota_K : \mathbb{R} &\longrightarrow V_h^0, \quad \iota_h(c) = c, \quad \begin{cases} \text{ran}(\iota_h) = \ker(\nabla), \\ \text{ran}(\nabla) = \ker(\nabla \times), \\ \text{ran}(\nabla \times) = \ker(\nabla \cdot), \\ \text{ran}(\nabla \cdot) = V_h^3, \end{cases} \\
0 \longrightarrow \mathbb{R} &\xrightarrow{c \mapsto c} V_h^0 \xrightarrow{\nabla} V_h^1 \xrightarrow{\nabla \times} V_h^2 \xrightarrow{\nabla \cdot} V_h^3 \longrightarrow 0 \quad \text{exact.}
\end{aligned}$$

□

5. STURM-LIOUVILLE SPECTRA

Separated realization, variational spectra, and conforming Ritz approximation are encoded by one weighted symmetric form [5, 6].

5.1. Self-adjoint realization.

Definition 5.1.1 (Regular separated Sturm–Liouville problem). Let $I = (0, L)$, $L > 0$, and assume [5, Chapters 1–4]

$$\begin{aligned} p &\in C^1([0, L]), & q, w &\in C([0, L]), & p &\geq p_0 > 0, & q &\geq 0, & w &\geq w_0 > 0 \quad \text{on } [0, L], \\ \{0, L\} &= \mathcal{E}_D \dot{\cup} \mathcal{E}_R, & \alpha_\xi &\geq 0 \quad (\xi \in \mathcal{E}_R), & \partial_n u(0) &= -u'(0), & \partial_n u(L) &= u'(L), & \mathcal{L}u &= -(pu')' + qu, \\ (\lambda, u) &\in \mathbb{R} \times (H^2(I) \setminus \{0\}), & & & \begin{cases} \mathcal{L}u = \lambda wu, & 0 < x < L, \\ u(\xi) = 0, & \xi \in \mathcal{E}_D, \\ p(\xi)\partial_n u(\xi) + \alpha_\xi u(\xi) = 0, & \xi \in \mathcal{E}_R. \end{cases} \end{aligned}$$

Definition 5.1.2 (Weighted form and weak eigenpair). Define

$$\begin{aligned} V &= \{v \in H^1(I) : v(\xi) = 0 \ (\xi \in \mathcal{E}_D)\}, & H_w &= L^2(I, w dx; \mathbb{R}), & m(u, v) &= \int_0^L w u v dx, \\ a_\alpha(u, v) &= \int_0^L (pu'v' + quv) dx + \sum_{\xi \in \mathcal{E}_R} \alpha_\xi u(\xi)v(\xi), & a_\alpha(u, v) &= \lambda m(u, v) \quad (v \in V), & u &\in V \setminus \{0\}. \end{aligned}$$

Proposition 5.1.3 (Weak–strong realization). For $f \in H_w$ and $u \in V$,

$$a_\alpha(u, v) = m(f, v) \quad (v \in V) \iff u \in H^2(I), \quad \begin{cases} \mathcal{L}u = wf, & 0 < x < L, \\ u(\xi) = 0, & \xi \in \mathcal{E}_D, \\ p(\xi)\partial_n u(\xi) + \alpha_\xi u(\xi) = 0, & \xi \in \mathcal{E}_R. \end{cases}$$

Proof. Let $u, v \in H^2(I)$; one integration by parts moves $\mathcal{L}$ onto v , the Robin data converts the boundary term into the α -sum, and endpoint hat functions recover each condition from the weak identity.

$$\begin{aligned} (\omega, \mathbf{G}) &= (v, pu' \mathbf{e}_1) \xrightarrow{1.2.6} \int_0^L (\mathcal{L}u)v dx = \int_0^L (pu'v' + quv) dx - [pu'v]_0^L, \\ p(\xi)\partial_n u(\xi) &= -\alpha_\xi u(\xi) \quad (\xi \in \mathcal{E}_R) \implies -[pu'v]_0^L = \sum_{\xi \in \mathcal{E}_R} \alpha_\xi u(\xi)v(\xi) \implies a_\alpha(u, v) = m(f, v) \quad (v \in V), \\ (u \in V, a_\alpha(u, v) &= m(f, v), H_0^1(I) \subseteq V) \implies \int_0^L pu'v' dx = \int_0^L (wf - qu)v dx \quad (v \in H_0^1(I)), \\ \int_0^L pu'v' dx &= -\langle (pu')', v \rangle_{\mathcal{D}'(I), \mathcal{D}(I)} \implies (pu')' = qu - wf \in L^2(I) \implies pu' \in H^1(I), \quad u \in H^2(I), \\ \eta_0(x) &:= 1 - \frac{x}{L}, \quad \eta_L(x) := \frac{x}{L}, \quad \eta_\xi \in V, \quad \eta_\xi(\zeta) = \delta_{\xi\zeta} \quad (\xi, \zeta \in \mathcal{E}_R), \\ v = \eta_\xi &\implies 0 = a_\alpha(u, \eta_\xi) - m(f, \eta_\xi) = p(\xi)\partial_n u(\xi) + \alpha_\xi u(\xi) \quad (\xi \in \mathcal{E}_R), \quad u \in V \xrightarrow{5.1.2} u|_{\mathcal{E}_D} = 0. \end{aligned}$$

□

Theorem 5.1.4 (Spectral resolution). Define $A = w^{-1}\mathcal{L}$ on H_w by

$$\mathcal{D}(A) = \{u \in H^2(I) \cap V : p(\xi)\partial_n u(\xi) + \alpha_\xi u(\xi) = 0 \ (\xi \in \mathcal{E}_R)\}, \quad Au = w^{-1}\mathcal{L}u.$$

Then A is nonnegative and self-adjoint with compact resolvent, and

$$\begin{aligned} 0 &\leq \lambda_1 < \lambda_2 < \dots, \quad \lambda_k \longrightarrow \infty, \quad m(\phi_j, \phi_k) = \delta_{jk}, \quad H_w = \overline{\text{span}\{\phi_k : k \geq 1\}}^{H_w}, \\ v \in V, \quad c_k &:= m(v, \phi_k) \implies m(v, v) = \sum_{k=1}^{\infty} |c_k|^2, \quad a_\alpha(v, v) = \sum_{k=1}^{\infty} \lambda_k |c_k|^2, \\ \lambda_1 &= 0 \iff (\mathcal{E}_D = \emptyset, \quad q \equiv 0, \quad \alpha_\xi = 0 \quad (\xi \in \mathcal{E}_R)). \end{aligned}$$

Proof. Let $b := a_\alpha + m$; the shifted form is bounded and coercive on V , so its solution operator inverts $A + I_{H_w}$, and compactness of $V \hookrightarrow H_w$ places that inverse under the spectral theorem of Appendix A.1.

$$b(v, v) \geq p_0 \|v'\|_{L^2(I)}^2 + w_0 \|v\|_{L^2(I)}^2 \geq c_b \|v\|_{H^1(I)}^2, \quad c_b := \min\{p_0, w_0\} > 0, \quad |b(u, v)| \leq C_b \|u\|_{H^1(I)} \|v\|_{H^1(I)},$$

$$\begin{aligned} (f \in H_w, V \text{ Hilbert, } b \text{ bounded coercive, } m(f, \cdot) \in V^*) &\xrightarrow{1.2.11} \exists! Tf \in V : b(Tf, v) = m(f, v) \quad (v \in V), \\ b(Tf, v) = m(f, v) &\implies a_\alpha(Tf, v) = m(f - Tf, v) \xrightarrow{5.1.3} Tf \in \mathcal{D}(A), \quad T = (A + I_{H_w})^{-1}, \quad \text{ran}(T) = \mathcal{D}(A), \\ m(Tf, g) = b(Tf, Tg) &= m(f, Tg), \quad m(Tf, f) = b(Tf, Tf) \geq 0, \quad Tf = 0 \implies f = 0 \implies \ker(T) = \{0\}, \\ V \hookrightarrow H_w \text{ compact} &\xrightarrow{A.1.5} T \text{ compact, positive, self-adjoint} \xrightarrow{A.1.3} T\phi_k = \mu_k \phi_k, \quad \mu_k \downarrow 0, \quad m(\phi_j, \phi_k) = \delta_{jk}, \\ H_w = \overline{\text{span}\{\phi_k\}}^{H_w}, \quad \lambda_k &:= \mu_k^{-1} - 1 \implies A\phi_k = \lambda_k \phi_k, \quad \lambda_k = a_\alpha(\phi_k, \phi_k) \geq 0, \quad \lambda_k \longrightarrow \infty. \end{aligned}$$

Expanding in the b -orthonormal frame $\eta_k = \sqrt{\mu_k} \phi_k$ gives Parseval in both forms, and a vanishing Wronskian forces simplicity.

$$b(\eta_j, \eta_k) = \delta_{jk}, \quad c_k = m(v, \phi_k) \implies b(v, v) = \sum_{k=1}^{\infty} (1 + \lambda_k) |c_k|^2 \implies a_\alpha(v, v) = \sum_{k=1}^{\infty} \lambda_k |c_k|^2,$$

$$\begin{aligned} Au = \lambda u, \quad Av = \lambda v, \quad W := p(uv' - u'v) &\implies W' = u(q - \lambda w)v - v(q - \lambda w)u = 0, \quad W(0) = 0 \implies v = cu, \\ \lambda_1 = 0 &\implies 0 = a_\alpha(\phi_1, \phi_1) \implies \phi_1 \equiv c \neq 0, \quad \mathcal{E}_D = \emptyset, \quad q \equiv 0, \quad \alpha_\xi = 0, \quad A1 = 0 \implies \lambda_1 = 0. \end{aligned}$$

□

Fact 5.1.5 (Sturm oscillation and interlacing). *For $k \geq 1$, ϕ_k has exactly $k - 1$ zeros in $(0, L)$, and the zeros of ϕ_k and ϕ_{k+1} interlace.*

Proof. [5, Chapters 4–5].

□

5.2. Min–max and Ritz approximation.

Theorem 5.2.1 (Rayleigh and min–max characterization). *For $0 \neq v \in V$, define*

$$\mathcal{R}_\alpha(v) = \frac{a_\alpha(v, v)}{m(v, v)}, \quad \lambda_k = \min_{\substack{S \subset V \\ \dim S = k}} \max_{0 \neq v \in S} \mathcal{R}_\alpha(v), \quad \lambda_1 = \min_{0 \neq v \in V} \mathcal{R}_\alpha(v) \quad (k \geq 1).$$

Proof. Let $v = \sum_j c_j \phi_j$ with $c_j = m(v, \phi_j)$; the quotient is a weighted average of the λ_j , bounded above on the first k modes and below on the m -orthogonal complement of the first $k - 1$.

$$v \in V \xrightarrow{5.1.4} m(v, v) = \sum_{j=1}^{\infty} |c_j|^2, \quad a_\alpha(v, v) = \sum_{j=1}^{\infty} \lambda_j |c_j|^2,$$

$$S_k := \text{span}\{\phi_1, \dots, \phi_k\}, \quad 0 \neq v \in S_k \implies \mathcal{R}_\alpha(v) = \frac{\sum_{j=1}^k \lambda_j |c_j|^2}{\sum_{j=1}^k |c_j|^2} \leq \lambda_k,$$

$$\dim S = k \implies S \cap \text{span}\{\phi_1, \dots, \phi_{k-1}\}^{\perp_m} \neq \{0\}, \quad 0 \neq v \in S \cap \text{span}\{\phi_1, \dots, \phi_{k-1}\}^{\perp_m} \implies \mathcal{R}_\alpha(v) \geq \lambda_k,$$

$$\lambda_k = \min_{\substack{S \subset V \\ \dim S = k}} \max_{0 \neq v \in S} \mathcal{R}_\alpha(v), \quad k = 1 \implies \lambda_1 = \min_{0 \neq v \in V} \mathcal{R}_\alpha(v).$$

□

Corollary 5.2.2 (Robin monotonicity). *Fix $p, q, w, \mathcal{E}_D, \mathcal{E}_R$ and let $0 \leq \alpha_\xi \leq \beta_\xi$ for $\xi \in \mathcal{E}_R$. Then*

$$\lambda_k(\alpha) \leq \lambda_k(\beta) \quad (k \geq 1), \quad \mathcal{E}_R = \{0, L\} \implies \lambda_k^N \leq \lambda_k^R(\alpha_0, \alpha_L) \leq \lambda_k^D,$$

and $\lambda_k^R(\alpha_0(t), \alpha_L(t)) \nearrow \lambda_k^D$ as $\min\{\alpha_0(t), \alpha_L(t)\} \rightarrow \infty$ [5, Chapter 4].

Proof. Let $0 \neq v \in V$; raising a Robin weight raises the numerator of the quotient pointwise, and the Dirichlet space is the subspace on which the boundary sum vanishes.

$$a_\beta(v, v) - a_\alpha(v, v) = \sum_{\xi \in \mathcal{E}_R} (\beta_\xi - \alpha_\xi) |v(\xi)|^2 \geq 0 \implies \mathcal{R}_\alpha(v) \leq \mathcal{R}_\beta(v) \xrightarrow{5.2.1} \lambda_k(\alpha) \leq \lambda_k(\beta),$$

$$V_D := \{v \in V : v(0) = v(L) = 0\} \subset V, \quad v \in V_D \implies \mathcal{R}_\alpha(v) = \mathcal{R}_0(v) \xrightarrow{5.2.1} \lambda_k^N \leq \lambda_k^R(\alpha_0, \alpha_L) \leq \lambda_k^D. \quad \square$$

Example 5.2.3 (Constant-coefficient separated spectra). For $\alpha_0, \alpha_L \geq 0$, consider

$$\begin{cases} -u'' = \lambda u, & 0 < x < L, \\ -u'(0) + \alpha_0 u(0) = 0, \\ u'(L) + \alpha_L u(L) = 0, \end{cases} \quad (p, q, w, \mathcal{E}_D, \mathcal{E}_R) = (1, 0, 1, \emptyset, \{0, L\}) \xrightarrow{5.1.4} \lambda \geq 0.$$

$$\lambda = \kappa^2 > 0 \implies u(x) = A \cos(\kappa x) + B \sin(\kappa x), \quad -u'(0) + \alpha_0 u(0) = 0 \implies A \neq 0, \quad B = \frac{\alpha_0}{\kappa} A,$$

$$u'(L) + \alpha_L u(L) = 0 \implies (\alpha_0 \alpha_L - \kappa^2) \sin(\kappa L) + \kappa(\alpha_0 + \alpha_L) \cos(\kappa L) = 0,$$

$$\lambda = 0 \implies u(x) = A + Bx, \quad B = \alpha_0 A, \quad A(\alpha_0 + \alpha_L + \alpha_0 \alpha_L L) = 0 \implies (0 \in \text{spec}) \iff \alpha_0 = \alpha_L = 0,$$

$$\lambda_k^N = \left(\frac{(k-1)\pi}{L} \right)^2, \quad \lambda_k^D = \left(\frac{k\pi}{L} \right)^2, \quad \lambda_k^{DN} = \left(\frac{(k-\frac{1}{2})\pi}{L} \right)^2 \xrightarrow{5.2.2} \lambda_k^N \leq \lambda_k^R \leq \lambda_k^D.$$

Proposition 5.2.4 (The P_1 matrix pencil). *Let*

$$0 = x_0 < x_1 < \dots < x_N = L, \quad T_r = [x_r, x_{r+1}], \quad h_r = x_{r+1} - x_r \quad (0 \leq r \leq N-1),$$

$$V_h = \{v_h \in C([0, L]) : v_h|_{T_r} \in \mathcal{P}_1(T_r), v_h(\xi) = 0 \ (\xi \in \mathcal{E}_D)\} \subset V, \quad \psi_j(z_i) = \delta_{ij},$$

$$K_{ij} = a_\alpha(\psi_j, \psi_i), \quad M_{ij} = m(\psi_j, \psi_i), \quad u_h = \sum_{j=1}^{N_h} U_j \psi_j,$$

$$a_\alpha(u_h, v_h) = \lambda_h m(u_h, v_h) \quad (v_h \in V_h) \iff K \cdot \mathbf{U} = \lambda_h M \cdot \mathbf{U}.$$

With $\mathbf{D}_r = [-h_r^{-1}, h_r^{-1}]^\top$ and $\mathbf{N}_r = [(x_{r+1} - x)/h_r, (x - x_r)/h_r]^\top$,

$$(p, q, w) \equiv (p_c, q_c, w_c) \implies K^{(r)} = \frac{p_c}{h_r} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} + \frac{q_c h_r}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad M^{(r)} = \frac{w_c h_r}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix},$$

$$M = M^\top \succ 0, \quad K = K^\top \succeq 0, \quad \lambda_k \leq \lambda_{k,h} \quad (1 \leq k \leq N_h).$$

Proof. Let $u_h = \sum_j U_j \psi_j$; expanding both forms in the nodal basis gives the pencil, elementwise integration assembles it, and the conforming inclusion inherits the min–max bound.

$$a_\alpha(u_h, \psi_i) = \sum_{j=1}^{N_h} a_\alpha(\psi_j, \psi_i) U_j, \quad m(u_h, \psi_i) = \sum_{j=1}^{N_h} m(\psi_j, \psi_i) U_j \xrightarrow{2.1.3} K \cdot \mathbf{U} = \lambda_h M \cdot \mathbf{U},$$

$$p \equiv p_c \implies \int_{x_r}^{x_{r+1}} p_c \mathbf{D}_r \cdot \mathbf{D}_r^\top dx = \frac{p_c}{h_r} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \quad s = \frac{x - x_r}{h_r} \implies \int_0^1 (1-s)^2 ds = \frac{1}{3}, \quad \int_0^1 s(1-s) ds = \frac{1}{6},$$

$$\int_{x_r}^{x_{r+1}} \mathbf{N}_r \cdot \mathbf{N}_r^\top dx = h_r \begin{bmatrix} \frac{1}{3} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{3} \end{bmatrix} = \frac{h_r}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad K = \sum_{r=0}^{N-1} E_r^\top K^{(r)} E_r + \sum_{\xi \in \mathcal{E}_R} \alpha_\xi \mathbf{e}_l(\xi) \mathbf{e}_l(\xi)^\top,$$

$$\mathbf{U}^\top \cdot M \cdot \mathbf{U} = m(u_h, u_h) > 0, \quad \mathbf{U}^\top \cdot K \cdot \mathbf{U} = a_\alpha(u_h, u_h) \geq 0 \implies M \succ 0, \quad K \succeq 0,$$

$$M = R^\top R, \quad \det R \neq 0 \implies [K \cdot \mathbf{U} = \lambda_h M \cdot \mathbf{U} \iff R^{-\top} K R^{-1} \cdot \mathbf{y} = \lambda_h \mathbf{y}], \quad \mathbf{y} = R \cdot \mathbf{U},$$

$$R^{-\top} K R^{-1} \text{ symmetric} \xrightarrow{A.1.2} \lambda_{k,h} = \min_{\substack{S_h \subset V_h \\ \dim S_h = k}} \max_{0 \neq v_h \in S_h} \mathcal{R}_\alpha(v_h), \quad V_h \subset V \xrightarrow{5.2.1} \lambda_k \leq \lambda_{k,h}.$$



Part 2. Appendix

APPENDIX A. FOUNDATIONAL TOOLS

A.1. Compact operators and the spectral theorem.

Definition A.1.1 (Adjoint, self-adjointness, compactness). Let H be a real Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and $S \in \mathcal{B}(H)$ [4, Appendix D].

$$\begin{aligned} \langle S^*x, y \rangle &= \langle x, Sy \rangle \quad (x, y \in H), \quad S = S^* \iff S \text{ self-adjoint}, \quad \langle Sx, x \rangle \geq 0 \quad (x \in H) \iff S \text{ positive}, \\ S \text{ compact} &\iff \left[(x_n)_{n \geq 1} \text{ bounded} \implies \exists n_1 < n_2 < \dots : (Sx_{n_\ell})_{\ell \geq 1} \text{ convergent} \right]. \end{aligned}$$

Proposition A.1.2 (Real spectrum and orthogonal eigenvectors). Let $S = S^* \in \mathcal{B}(H)$. Then every eigenvalue of S is real, eigenvectors for distinct eigenvalues are orthogonal, and

$$S \text{ positive}, \quad Sx = \mu x, \quad x \neq 0 \implies \mu \geq 0.$$

Proof. Let $Sx = \mu x$ and $Sy = \nu y$ with $x, y \neq 0$; self-adjointness evaluates one pairing two ways, and the two evaluations differ by the factor $\mu - \nu$.

$$\begin{aligned} \mu \langle x, x \rangle &= \langle Sx, x \rangle = \langle x, Sx \rangle = \mu \langle x, x \rangle \implies \mu \in \mathbb{R}, \quad S \text{ positive} \implies \mu \langle x, x \rangle = \langle Sx, x \rangle \geq 0 \implies \mu \geq 0, \\ \mu \langle x, y \rangle &= \langle Sx, y \rangle = \langle x, Sy \rangle = \nu \langle x, y \rangle \implies (\mu - \nu) \langle x, y \rangle = 0 \implies \left[\mu \neq \nu \implies \langle x, y \rangle = 0 \right]. \end{aligned}$$

□

Theorem A.1.3 (Spectral theorem for compact self-adjoint operators). Let $S = S^* \in \mathcal{B}(H)$ be compact, positive, and injective on a separable H . Then [4, Appendices D.5–D.6]

$$\begin{aligned} S\phi_k &= \mu_k \phi_k, \quad \mu_1 \geq \mu_2 \geq \dots > 0, \quad \lim_{k \rightarrow \infty} \mu_k = 0, \quad \langle \phi_j, \phi_k \rangle = \delta_{jk}, \quad H = \overline{\text{span}\{\phi_k : k \geq 1\}}^H, \\ x \in H, \quad c_k &:= \langle x, \phi_k \rangle \implies x = \sum_{k=1}^{\infty} c_k \phi_k, \quad Sx = \sum_{k=1}^{\infty} \mu_k c_k \phi_k, \quad \mu_1 = \max_{x \neq 0} \frac{\langle Sx, x \rangle}{\langle x, x \rangle}. \end{aligned}$$

Proof. [4, Appendices D.5–D.6]; the eigenvalues are real and nonnegative and the eigenvectors orthogonal by Proposition A.1.2, and $\ker(S) = \{0\}$ makes the eigenvectors total. □

Example A.1.4 (A diagonal model). Let $H = \ell^2(\mathbb{R})$, let $(\mu_k)_{k \geq 1}$ satisfy $\mu_k > 0$ and $\mu_k \rightarrow 0$, and set $Sx = (\mu_k x_k)_{k \geq 1}$.

$$\begin{aligned} \langle Sx, y \rangle &= \sum_{k=1}^{\infty} \mu_k x_k y_k = \langle x, Sy \rangle \implies S = S^*, \quad \langle Sx, x \rangle = \sum_{k=1}^{\infty} \mu_k |x_k|^2 \geq 0 \implies S \text{ positive}, \\ S_N x &:= (\mu_1 x_1, \dots, \mu_N x_N, 0, 0, \dots) \implies \|S - S_N\| \leq \sup_{k > N} \mu_k \longrightarrow 0 \implies S \text{ compact}, \\ S e_k &= \mu_k e_k, \quad \langle e_j, e_k \rangle = \delta_{jk}, \quad \ker(S) = \{0\} \xrightarrow{\text{A.1.3}} H = \overline{\text{span}\{e_k : k \geq 1\}}^H. \end{aligned}$$

Fact A.1.5 (Rellich–Kondrachov on an interval). Let $I = (0, L)$ be bounded and let $V \subseteq H^1(I)$ be a closed subspace. Then the inclusion is compact,

$$V \hookrightarrow L^2(I) \text{ compact}, \quad \|v\|_{L^\infty(I)} \leq C \|v\|_{H^1(I)} \quad (v \in H^1(I)).$$

Proof. [4, Section 5.7]. □

Example A.1.6 (Compactness fails without the bounded domain). Let $I = \mathbb{R}$, let $0 \neq \varphi \in C_c^\infty(\mathbb{R})$, and set $\varphi_n(x) := \varphi(x - n)$.

$$\begin{aligned} \|\varphi_n\|_{H^1(\mathbb{R})} &= \|\varphi\|_{H^1(\mathbb{R})} \implies (\varphi_n)_{n \geq 1} \text{ bounded in } H^1(\mathbb{R}), \quad n \neq k, \quad |n - k| > \text{diam supp}(\varphi) \implies \varphi_n \varphi_k \equiv 0, \\ \|\varphi_n - \varphi_k\|_{L^2(\mathbb{R})}^2 &= \|\varphi_n\|_{L^2(\mathbb{R})}^2 + \|\varphi_k\|_{L^2(\mathbb{R})}^2 = 2\|\varphi\|_{L^2(\mathbb{R})}^2 > 0 \implies (\varphi_n)_{n \geq 1} \text{ has no } L^2\text{-convergent subsequence.} \end{aligned}$$

Theorem 5.1.4 applies Fact A.1.5 to the resolvent $T = (A + I_{H_w})^{-1}$ and then Theorem A.1.3 to T , and Proposition 5.2.4 applies Proposition A.1.2 to the congruent matrix $R^{-T}KR^{-1}$.

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